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TEST BOOKLET SERIES

**Test Code:
2205****TEST BOOKLET
GENERAL STUDIES
TEST – 5****A****Time Allowed : Two Hours****Maximum Marks : 200****INSTRUCTIONS**

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS TEST BOOKLET *DOES NOT* HAVE ANY UNPRINTED OR TORN OR MISSING PAGES OR ITEMS, ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET.
2. ENCODE CLEARLY THE TEST BOOKLET SERIES CODE **A, B, C** OR **D** AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.
3. You have to enter your Roll Number on the Test Booklet in the Box provided alongside this line. *DO NOT* write *anything else* on the Test Booklet.
4. This Test Booklet contains 100 items (questions). Each question is printed both in **Hindi** and **English**. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response, mark the response which you consider the best. In any case, choose **ONLY ONE** response for each item.
5. You have to mark all your responses **ONLY** on the separate Answer Sheet provided. See directions in the Answer sheet.
6. All items carry equal marks.
7. Before you proceed to mark in the Answer Sheet the response to various items in the Test Booklet, you have to fill in some particulars in the Answer Sheet as per instructions sent to you with your admission certificate.
8. After you have completed filling in all your responses on the Answer Sheet and examination has concluded, you should hand over to the Invigilator *only the Answer Sheet*. You are permitted to take away with you the Test Booklet.
9. Sheets for rough work are appended in the Test Booklet at the end.
10. **Penalty for wrong answers:**
THERE WILL BE PENALTY FOR WRONG ANSWERS MARKED BY A CANDIDATE IN THE OBJECTIVE TYPE QUESTION PAPERS.
 - (i) There are four alternatives for the answer to every question. For each question for which a wrong answer has been given by the candidate, **One-Third** of the marks assigned to that question will be deducted as penalty.
 - (ii) If a candidate gives more than one answer, it will be treated as a **wrong answer** even if one of the given answers happens to be correct and there will be same penalty as above to that question.
 - (iii) If a question is left blank, i.e., no answer is given by the candidate, there will be no **penalty** for that question.

@UPSC_SUCCESS_TIME1**DO NOT OPEN THIS TEST BOOKLET UNTIL YOU ARE ASKED TO DO SO***Note:* Hindi version of the instructions is printed on the front cover of this Booklet.

1. In India, majority of the copper reserves are found in which of the following range?

- (a) Vindhya range (b) Satpura range
(c) Purvanchal range (d) Aravali range

2. Consider the following statements regarding World Heritage Sites:

1. The World Heritage Centre is the Secretariat to the 1972 Convention.
2. It is voluntary to put any monument/site on the Tentative List (TL) before applying for recognition.
3. Ghats of Varanasi has been recently included in the tentative list of UNESCO's world heritage sites.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

3. Consider the following characteristics of a soil type:

1. It is found in areas with high temperature and high rainfall.
2. Intensively leached due to rains.
3. Not suitable for cultivation unless fertilizers and manures are applied.

Identify the soil mentioned in the above statements:

- (a) Red and Yellow Soil (b) Laterite Soil
(c) Black Soil (d) Arid Soil

4. With reference to Rabindranath Tagore, consider the following statements:

1. He established the Vishwa-Bharati University in 1921 to challenge conventional education.
2. He was the first non-European to receive the Nobel Prize.
3. He was awarded a knighthood by the British King George V in 1919 after the Jallianwalla Bagh massacre.

Which of the statements given above is/are **not** correct?

- (a) 1 and 2 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

5. Which of the following are the objectives of the SWAMIH Fund?

- (a) To complete construction of stalled, affordable and mid-income category housing projects.
- (b) To address the housing requirement in tribal areas and islands.
- (c) To build schools for Scheduled Tribe children in remote areas to impart quality education.
- (d) To support children who have lost both parents or their lone surviving parent or their legal guardian or adoptive parent due to COVID-19.

6. Which of the following statements is/are correct?

1. Chhattisgarh shares its boundary with Bihar.
2. Imphal is located north of Kohima.

Select the correct answer using the codes given below:

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

7. With reference to the Bugyals, consider the following statements:

1. The entire alpine zone of Uttarakhand comes under the category of Bugyal.
2. Bugyals are the repositories of a large number of medicinal and aromatic plants which are used in local medicine and in herbal industries.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

8. Consider the following statements regarding the Competition Commission of India (CCI):

1. It comes under the Ministry of Commerce and Industry.
2. It aims to eliminate practices having adverse effects on competition, promote and sustain competition.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

9. Consider the following statements:
1. Meghalaya is also known as Molasses basin.
 2. Barak River is a tributary of Brahmaputra.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
10. With reference to the OSIRIS-REx Mission, consider the following statements:
1. The OSIRIS-REx is NASA's first mission to visit a near-Earth asteroid.
 2. The OSIRIS-REx spacecraft was launched in 2016 for the journey to the Bennu asteroid.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
11. Consider the following statements:
1. In India, the area under tropical dry deciduous forests is the highest relative to any other type of forest cover.
 2. Littoral and swamp forests are found in and around deltas and estuaries only they do not occur around creeks.
- Which of the statements given above is/are correct?
- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
12. Consider the following statements:
1. Bardoli Chheerha are local storms in Assam and are useful for Jute cultivation.
 2. Pre-monsoon showers in Kerala, often referred to as 'mango showers' help in the early ripening of mangoes.
 3. Norwesters are also known in local nomenclature as 'Kalbaisakhi'.
- Which of the above statements is/are correct?
- (a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2, and 3
13. With reference to the National Financial Reporting Authority (NFRA), consider the following statements:
1. NFRA was constituted under the Companies Act, 2013.
 2. The NFRA account is monitored by the Comptroller and Auditor General of India.
 3. NFRA cannot undertake an investigation of companies that are listed on any stock exchange.
- Which of the statements given above are correct?
- (a) 2 and 3 only (b) 1 and 2 only
(c) 1 and 3 only (d) 1, 2 and 3
14. The Nine Degree Channel separates which of the following?
- (a) Minicoy and Lakshadweep
(b) Agatti and Kavaratti
(c) Kalpeni and Minicoy
(d) Bitra and Androth
15. Consider the following states:
1. Odisha 2. Jharkhand
 3. West Bengal 4. Chattisgarh
- In terms of the percentage of coal reserves, which one of the following is the correct descending order?
- (a) 2-1-4-3 (b) 2-4-1-3
(c) 4-2-3-1 (d) 4-2-1-3
16. Match the following:
- | GM Crop | Modified crop Traits |
|----------------------|----------------------------------|
| 1. BT cotton | (i) Fortification with Vitamin A |
| 2. Mustard DMH-11 | (ii) Increased shelf life |
| 3. Golden rice | (iii) Resistance to bollworm |
| 4. Flavr Savr tomato | (iv) Herbicide tolerant crop |
- Select the correct answer using the code given below:
- (a) 1-iii, 2-ii, 3-i, 4-iv
(b) 1-iii, 2-iv, 3-i, 4-ii
(c) 1-iv, 2-i, 3-ii, 4-iii
(d) 1-iii, 2-i, 3-ii, 4-iv

17. The fertile alluvial soils are becoming saline in areas of intensive cultivation. In this context, consider the following statements:

1. Excessive irrigation with dry climatic conditions promotes capillary action, resulting in deposition of salt on the soil.
2. Farmers are advised to add gypsum to solve the problem of salinity.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

18. With reference to the Civil Services in India, consider the following statements:

1. Article 311 of the Indian Constitution provides protection to Civil servants of the Union and State along with defence personnel.
2. The Civil servants of the Union and State work at the pleasure of the President of India.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

19. With reference to Montane forests, consider the following statements:

1. The temperate forests in Himalayas are called Sholas.
2. Chir pine, Deodar and Birch are the important species in Himalayan region.
3. Mosses and lichens form part of the tundra vegetation.
4. The southern slopes of Himalayas carry a thicker vegetation cover as compared to the northern slopes.

Which of the statements given above are correct?

- (a) 1, 3 and 4 only (b) 2, 3 and 4 only
(c) 1, 2 and 4 only (d) 1, 2, 3 and 4

20. Ritchie's archipelago and Labyrinth islands are part of which Island group?

- (a) Bay of Bengal Island group
(b) Arabian Sea Island group

- (c) Katchatheevu Islands
(d) None of the above

21. Which of the following statements is **not** correct?

- (a) The prison is a State subject under the Seventh Schedule of the Indian Constitution.
(b) The Ministry of Home Affairs is responsible for providing regular guidance to States and UTs on issues concerning prisons and prison inmates.
(c) Article 22 of the Indian Constitution allows for preventive detention and restriction on personal liberty for reasons of state security and public order.
(d) Article 23 of the Indian Constitution ensures free legal aid to promote justice on a basis of equal opportunity.

22. Arrange the following hills of North-Eastern India in their North to South order:

1. Patkai Bum Hills 2. Mishmi Hills
3. Mizo Hills 4. Khasi Hills

Select the correct answer using the codes given below:

- (a) 1-2-4-3 (b) 2-1-4-3
(c) 2-1-3-4 (d) 1-2-3-4

23. With reference to the Pradhan Mantri Awaas Yojana-Gramin, consider the following statements:

1. National Family Health Survey (NFHS-5) data will be used for the identification of beneficiaries.
2. It aims to provide a pucca house to all rural families irrespective of their status by the end of March 2022.
3. It comes under the Ministry of Rural Development.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 1 and 3 only
(c) 3 only
(d) 1, 2 and 3

24. Consider the following statements about the Burhi Dihing river:

1. Burhi Dihing river is a tributary of Brahmaputra river.
2. It originates from Patkai Hills.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

25. Consider the following statements regarding Peninsular Plateau:

1. It is made up of igneous and metamorphic rocks.
2. It is an unstable zone highly prone to earthquakes.
3. The plateau has broad and shallow valleys and rounded hills.

Which of the following statements are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

26. With reference to Nuclear Minerals, consider the following statements:

1. Uranium ores are not found in India.
2. India is rich in Thorium reserves.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

27. Consider the following statements:

1. Tropical Cyclones are characterized by low atmospheric pressure, high winds, and heavy rain.
2. The name Cyclone Tauktae was given by Thailand.
3. The India Meteorological Department is mandated to issue advisories and name tropical cyclones in the North Indian Ocean Region.

Which of the statements given above is/are correct?

- (a) 1 and 3 only
(b) 2 and 3 only
(c) 2 only
(d) 1, 2 and 3

28. Consider the following pairs:

Rivers	Originates from
1. Narmada	: Amarkantak Hills
2. Kaveri	: Mahabaleshwar
3. Krishna	: Brahmagiri Range

Which of the pairs given above is/are correctly matched?

- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

29. Consider the following statements:

1. River Jhelum forms meanders in its youth stage.
2. The meanders in Jhelum river are caused by the local base level provided by the lakes.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

30. With reference to ferrous minerals, consider the following statements:

1. Iron ore, Manganese and Chromite are Ferrous minerals.
2. Haematite and Magnetite are the most prominent iron ores found in India.
3. Dalli and Rajhara in Durg are the important mines of iron ore in the country.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

31. Consider the following statements:

1. Humid climatic conditions are highly suitable for growing of cotton.
2. Black soil is the most suitable soil for the cultivation of cotton.
3. Production of cotton is the largest in the state of Maharashtra.

Which of the statements given above are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

32. Consider the following statement regarding the 'Particularly Vulnerable Tribal Groups (PVTGs)'
1. The Dhebar Commission created Primitive Tribal Groups (PTGs) as a separate category in 1973.
 2. The highest number of PVTGs are found in Andaman & Nicobar Islands.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

33. Veliconda group of low hills is a structural part of
- (a) Nilgiri Hills (b) Western Ghats
(c) Eastern Ghats (d) Cardamom Hills

34. Consider the following statement regarding "Banni Grassland":

1. It is situated in the Indian state of Gujarat.
2. It is the largest tropical grassland in Asia.
3. Rabari tribe is the nomadic native of this grassland.

Which of the statements given above is/are correct?

- (a) 1 only (b) 1 and 2 only
(c) 2 and 3 only (d) 3 only

35. Every year, 22nd May is celebrated as International Day for Biological Diversity, consider the following statements regarding that:

1. It aims to increase understanding and awareness of biodiversity issues.
2. The theme for 2021 is 'Our solutions are in nature'.
3. The year 2021-2030 is declared as the UN Decade of Ocean Science for Sustainable Development and the UN Decade on Ecosystem Restoration.

Which of the statements given above is/are correct?

- (a) 1 and 3 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

36. Which of the following migration streams is usually dominated by female migrants in India?

- (a) Rural-rural (b) Rural-urban
(c) Urban-rural (d) Urban-urban

37. According to ICAR classification which types of soils are found in India?

1. Inceptisols 2. Entisols
3. Alfisols 4. Vertisols

Select the correct answer using the code given below:

- (a) 1 and 3 only (b) 2 and 3 only
(c) 1 and 2 only (d) 1, 2, 3 and 4

38. Consider the following statements:

1. The Juvenile Justice Act, 2015 provides for the process of taking care of children who are orphaned.
2. India has not ratified the Hague Convention on Inter-country Adoptions, 1993.
3. The State Commission for Protection of Child Rights is the nodal agency for adoption in India.

Which of the statements given above is/are **not** correct:

- (a) 1 and 2 only (b) 2 and 3 only
(c) 3 only (d) 1 and 3 only

39. Arrange the following from North to South:

1. Palani Hills
2. Cardamom Hills
3. Nilgiri Hills

Select the correct answer using the code given below:

- (a) 1-2-3 (b) 3-2-1
(c) 3-1-2 (d) 2-1-3

40. Which of the following conditions are necessary for delta formation?

1. A shallow shore around the river mouth.
2. The flow of river must be low near the mouth.
3. Tidal waves near the mouth must be strong.

Select the correct answer using the code given below:

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

41. Recently seen in the news, CoviSelf is

- (a) India's first self-use Rapid Antigen Test (RAT)
(b) Advanced Chaff Technology developed by DRDO
(c) Application to monitor the home quarantined COVID-19 patient
(d) A medicine on COVID-19 discovered by Defence Research and Development Organisation (DRDO)

42. Which of the following pairs are correctly matched?

Tree Species	Type of Forest
1. Juniper	– Montane
2. Shisham	– Tropical Deciduous
3. Mahogany	– Tropical Evergreen

Select the correct answer using the code given below:

- (a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

43. Which of the following states of India share a border with Myanmar?

1. Mizoram 2. Arunachal Pradesh
3. Tripura 4. Manipur

Select the correct answer using the code given below:

- (a) 1, 2 and 3 only (b) 1, 3 and 4 only
(c) 2, 3 and 4 only (d) 1, 2 and 4 only

44. Consider the following statements regarding International Museum Day:

1. It is observed to raise awareness among people about museums.
2. The Day was instituted in 1977 by the International Council of Museums (ICOM).
3. All Museums in India are administered by the Ministry of Culture.

Which of the statements given above is/are correct?

- (a) 2 only (b) 1 and 2 only
(c) 1 only (d) 1, 2 and 3

45. Which of the following rivers drain in the Bay of Bengal?

1. Mahi 2. Penner
3. Subarnarekha 4. Tapti

Select the correct answer using the code given below:

- (a) 1 and 4 only (b) 2 and 3 only
(c) 1, 2 and 3 only (d) 3 and 4 only

46. Consider the following statements:

1. Percentage of area prone to flooding in India is the highest in the state of Bihar.
2. In some parts of West Bengal, floods are caused by tidal effects.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

47. Consider the following statements:

1. The Legislative Council of a State shall not have more than one-third of the total strength of the State Assembly.
2. The members of the Legislative Council cannot vote in elections for the President and Vice President.
3. In the election of members of the Legislative Council, there is a provision for a special electorate comprising sitting members of local governments.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 2 and 3 only
(c) 2 only (d) 1, 2 and 3

48. Which of the following pairs is/are correctly matched?

Port	Feature
1. Marmagao Port	– Zuari Estuary
2. Kochchi Port	– Landlocked Harbour
3. Visakhapatnam	– Vembanad Kayal

Select the correct answer using the code given below:

- (a) 1 only (b) 2 and 3 only
(c) 1 and 2 only (d) 1 and 3 only

49. Consider the following statements:

1. UNESCO is responsible for establishing professional and ethical standards for museum related activities worldwide.
2. Article 49 of the Indian Constitution directs the state to protect objects of artistic or historic interest that are of national importance.
3. The National Portal and Digital Repository for Indian Museums is an initiative of the Ministry of Culture for the digitisation of the collections of the Museums.

Which of the statements given above is/are correct?

- (a) 1 and 2 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

50. Consider the following statements:

1. Majority of glaciers in India are situated in the Pir Panjal range.
2. Gangotri glacier is located in the Kumaon-Garhwal region of Himalayas.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

51. With reference to the Black Soil, consider the following statements:

1. It is also known as the Regur Soil.
2. During dry seasons, black soil develops wide cracks.
3. Moisture in black soil evaporates in a short time.

Which of the statements given above is/are correct?

- (a) 1 and 3 only (b) 1 only
(c) 2 and 3 only (d) 1 and 2 only

52. Consider the following statements:

1. Through Project Elephant, the government provides financial and technical support to major elephant range states in the country.
2. Assam has the highest number of elephants.
3. Asian elephants are classified as critically endangered in the IUCN Red List.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 and 3 only
(c) 1 and 2 only (d) 1, 2 and 3

53. Consider the following statements about Biomes:

1. Taiga biome is devoid of trees and the flora here includes lichens and mosses.
2. Low rainfall and wild fires disrupt the maintenance of grasslands.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

54. Which of the following Himalayan mountain passes are correctly matched?

Mountain Pass	State
1. Zojila Pass	– Arunachal Pradesh
2. Nathu la Pass	– Sikkim
3. Rohtang la Pass	– Himachal Pradesh
4. Lipulekh Pass	– Jammu and Kashmir

Select the correct answer using the code given below:

- (a) 1 and 2 only
(b) 2 and 3 only
(c) 3 and 4 only
(d) 1, 2, 3 and 4

55. Consider the following statements regarding Crime and Criminal Tracking Network and Systems (CCTNS):

1. It is developed by the National Crime Records Bureau (NCRB).
2. It is a Mission Mode Project under the National e-Governance Plan of the Ministry of Science and Technology.
3. CCTNS aims to make the Police functioning citizen-friendly.

Which of the statements given above are correct?

- (a) 2 and 3 only
(b) 1 and 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

56. With reference to the Panch Prayag, which of the following pairs is/are correctly matched:

Confluence Rivers

- | | | |
|------------------|---|----------------------------|
| 1. Vishnu Prayag | – | Alaknanda and Bhagirathi |
| 2. Nand Prayag | – | Alaknanda and Nandakini |
| 3. Karna Prayag | – | Alaknanda and Mandakini |
| 4. Rudra Prayag | – | Alaknanda and Pindar |
| 5. Devprayag | – | Alaknanda and Dhauli Ganga |

Select the correct answer using the codes given below:

- | | |
|---------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) 2, 4 and 5 only | (d) 1, 2 and 3 only |

57. Consider the following statements:

1. Rainforest ecosystems can be found on all the continents except Antarctica.
2. Deciduous forests are Earth's oldest surviving ecosystems.

Which of the statements given above is/are correct?

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

58. Consider the following statements:

1. Di-Ammonium Phosphate (DAP) fertiliser is the second most commonly used fertilizer in India after urea.
2. All urea-based fertilisers are regulated under the Nutrient Based Subsidy Scheme.

Which of the statements given above is/are correct?

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

59. Consider the following statements regarding Harmonised Master List of Infrastructure sub-sectors:

1. The list is notified by the Ministry of Commerce and Industries.
2. Social and Commercial Infrastructure is one of the categories included in the list.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

60. With reference to Electronic Way (E-Way) Bill, consider the following statements:

1. It is mandatory for inter-state transportation of goods valued over Rupee 50,000.
2. It is a tool to track the movement of goods and check tax evasion.

Which of the statements given above is/are correct?

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

61. With reference to the classification of irrigation, consider the following statements:

1. The protective irrigation acts as a supplement to the water provided by rainfall.
2. In productive irrigation, water input per unit area is higher than the protective irrigation.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

62. Consider the following statements regarding the Transgender Persons (Protection of Rights) Act, 2019:

1. The Act declared transgender persons as the third gender.
2. The Act established the National Council for Transgender persons.

Which of the statements given above is/are **not** correct?

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

63. With reference to Tropical Cyclones, consider the following statements:

1. The eye of the Cyclone is Usually the area which has the highest turbulence.
2. Pressure is found to be very high at the centre of the cyclones.
3. Vertical clouds with strong winds surround the area of the eye of the tropical cyclone.

Which of the statements given above is correct?

- (a) 1 only (b) 2 only
(c) 3 only (d) None of the above

64. With reference to the Census of 2011, consider the following statements:

1. Uttar Pradesh showed the highest percentage change in Population and decadal change by residence from 2001 to 2011.
2. No state and Union territory showed negative growth in Population between 2001 and 2011.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

65. With reference to Regulations Review Authority, consider the following statements:

1. The Ministry of Finance has set up the Regulations Review Authority.
2. It will focus on streamlining regulatory instructions and reducing the compliance burden of the regulated entities by simplifying procedures.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

66. Consider the following statements regarding the city of Kanchipuram:

1. The city is situated on the banks of the Kaveri river.
2. The city was the capital of the Pallava dynasty in the 6th and 7th centuries.
3. The temples of Kanchipuram were declared as a World Heritage Site by UNESCO in 2011.

Which of the statements given above is/are **not** correct?

- (a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 3 only

67. Consider the following statements:

1. If farmer specializes in cash crops only, than it is called truck farming.
2. Kolkhoz is a type of collective farming prevalent in Soviet Union.
3. Footloose industries are not dependent on any specific raw material.

Which of the statements given above are correct?

- (a) 2 and 3 only (b) 1 and 2 only
(c) 1, 2 and 3 (d) 1 and 3 only

68. Consider the following pairs:

Climate	Region
1. Monsoon with dry winter	– West coast of India (South of Goa)
2. Tropical Savannah	– North-western Gujarat, some parts of Western Rajasthan and Punjab
3. Monsoon with dry summer	– Coromandel Coast of Tamil Nadu
4. Cold Humid winter with short summer	– Himachal Pradesh and Uttarakhand

Which of the pairs given above is/are correctly matched?

- (a) 1 only (b) 3 only
(c) 2 and 4 only (d) 1 and 3 only

69. Consider the following pairs:

Movements	States
1. Bishnoi Movement	– Rajasthan
2. Appiko Movement	– Assam
3. Chipko Movement	– Uttarakhand
4. Jungle Bachao Andolan	– West Bengal

Which of the pairs given above is/are correctly matched?

- (a) 1 only (b) 1 and 4 only
(c) 1 and 3 only (d) 1, 2 and 4

70. Arrange the following mountain ranges in the North to South order:

1. Karakoram 2. Zaskar
3. Pir Panjal 4. Ladakh

Select the correct answer using the codes given below:

- (a) 3-1-2-4 (b) 2-1-3-4
(c) 1-4-2-3 (d) 1-2-3-4

71. The “India Energy Outlook 2021 Report” is released by which of the following?

- (a) Ministry of New and Renewable Energy
(b) NITI Aayog
(c) International Energy Agency
(d) Ministry of Power

72. The Eastern Ghats are spread over which of the following Indian states?

1. West Bengal 2. Telangana
3. Andhra Pradesh 4. Kerala

Select the correct answer using the code given below:

- (a) 1, 3 and 4 only (b) 2 and 3 only
(c) 1, 2 and 3 only (d) 2, 3 and 4 only

73. With reference to the Van Gujjars, consider the following statements:

1. These are the forest-dwelling nomadic communities inhabiting only in the state of Uttarakhand.
2. They pursue seasonal migration from the Terai-Bhabar and Siwalik region of Uttarakhand to the higher bugyals in the Western Himalayas.

Which of the statements given above is/are **not** correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

74. Which of the following rivers surround the largest riverine island of Majuli?

1. Kherkutia Suti 2. Brahmaputra
3. Lohit 4. Subansiri

Select the correct answer using the codes given below:

- (a) 1, 2 and 4 only (b) 1, 2 and 3 only
(c) 2, 3 and 4 only (d) 1, 2, 3 and 4

75. Which of the following Ramsar sites in India is a freshwater body?

- (a) Lonar Lake (b) Chilika Lake
(c) Kolleru Lake (d) Pangong Tso

76. Consider the following statements regarding Pathogens:

1. Zoonotic pathogen refers to pathogens transmitted naturally between animals and humans.
2. Leprosy and Cholera are transmitted by pathogenic bacteria.
3. The BioHub facility is launched by the G20 countries to facilitate pathogens sharing and preparedness against them.

Which of the statements given above is/are **not** correct?

- (a) 1 and 2 only (b) 2 only
(c) 3 only (d) 2 and 3 only

77. Consider the following statements:

1. Khasi and Jaintia hills are part of the peninsular plateau.
2. The pass located at the tri-junction of India, China and Myanmar is Diphu pass.
3. Anthracite coal is found in Jammu and Kashmir.

Which of the statements given above is/are correct?

- (a) 2 only
(b) 1, 2 and 3
(c) 1 and 3 only
(d) 3 only

78. Consider the following statements:

1. White Fungus is a fungal infection caused by a yeast called Candida.
2. Candida normally lives on the skin and inside the body without causing any problems.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

79. Consider the following statements:

1. The retreating southwest monsoon season is marked by clear skies and rise in temperature in most parts of India.
2. An increase in the night temperature indicates an advance in the arrival of western cyclonic disturbances.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

80. With reference to Lunar Eclipse, consider the following statements:

1. Penumbra is the part of the Earth's shadow where direct light from the Sun is totally blocked out by the Earth.
2. Umbra is the portion where not all of the light from the Sun is blocked out.

Which of the statements given above is/are **not** correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

81. Consider the following statements:

1. Culturable wasteland is any land left uncultivated for one year.
2. Fallow other than Current Fallow is the land left uncultivated for less than one year.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

82. Consider the following statements:

1. Narmada is the largest west flowing river of peninsular India.
2. Cauvery river has the largest river basin of peninsular India.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

83. With reference to Mid-Day-Meal Scheme, consider the following statements:

1. It is a centrally sponsored scheme which was launched in 1995.
2. It aimed to attain the goal of universalization of primary education.
3. The nodal ministry for its implication is the Ministry of Health and Family Welfare.

Which of the statements given above is/are correct?

- (a) 2 and 3 only (b) 2 only
(c) 1 and 2 only (d) 1, 2 and 3

84. Which of the following factors are responsible for making Punjab and Haryana one of the largest rice producing states?

1. Genetically improved seed varieties.
2. High uses of pesticides and fertilizers.
3. Dry climatic conditions

Select the correct answer using the codes given below:

- (a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

85. With reference to the Dharwar Rock System, consider the following statements:

1. They are mainly found in the Chotanagpur plateau and Meghalaya.
2. They are rich in metallic minerals such as iron, manganese, gold, copper.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

86. The primary goal of Pradhan Mantri Swasthya Suraksha Yojana (PMSSY) is to:

- (a) Protect poor and vulnerable families against financial risk arising out of catastrophic health episodes
- (b) Correct regional imbalances in the availability of healthcare services
- (c) Right to healthcare to all adult citizens
- (d) Provide medicines to poor people suffering from cardiovascular diseases or cancer

87. Which of the following is/are correct about Bauxite:

- 1. It occurs in laterite rocks.
- 2. It is important in the aircraft manufacturing industry.
- 3. Odisha is its largest producer.

Select the correct answer using the code given below:

- (a) 2 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

88. Consider the following statements regarding the Non-Aligned Movement (NAM):

- 1. It was formed during the Cold War as an organisation of States that did not seek to formally align themselves with either the USA or the Soviet Union.
- 2. The founding principles of the NAM were adopted in the Havana Declaration.
- 3. India was one of the leading countries of NAM under the Prime Ministership of Jawaharlal Nehru.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

89. Consider the following statements:

- 1. In the months of March to May Northern India experiences thunderstorms which are usually accompanied by hail.
- 2. Western Disturbances do not cause hail storms in the winter season.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

90. Which of the following cities is closest to the Tropic of Cancer?

- (a) Gandhinagar
- (b) Ranchi
- (c) Bhopal
- (d) Kolkata

91. Consider the following statements:

- 1. The World Press Freedom Day is organised by the UN Educational, Scientific and Cultural Organization (UNESCO).
- 2. The Windhoek Declaration is a benchmark for ensuring press freedom around the world.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

92. With reference to National Programme on Advanced Chemistry Cell (ACC) Battery Storage, consider the following statements:

- 1. It comes under the Ministry of Heavy Industries & Public Enterprises.
- 2. It aims to set up a 50-gigawatt hour (GWh) manufacturing capacity for ACC batteries.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

93. Consider the following statements:

- 1. Physiological density is the ratio of total population and sum of net cultivated and residential area.
- 2. Agricultural density is the ratio of total agricultural population and net cultivable area.
- 3. The definition of Agricultural population includes cultivators and their family members only.
- 4. The Population doubling time is the time taken by any population to double itself at its current decadal growth rate.

Which of the statements given above are **not** correct?

- (a) 1, 2 and 3 only (b) 1, 3 and 4 only
(c) 3 and 4 only (d) 1, 2, 3 and 4

94. Consider the following statements

1. The National Small Savings Fund (NSSF) was established within the Consolidated Fund of India.
2. The rules to administer NSSF are framed by the President of India under Article 283 (1) of the Indian Constitution.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

95. With reference to the Northern Plains, arrange the following in the South to North orientation:

1. Terai 2. Bhabar
3. Bangar 4. Khadar

Select the correct answer using the code given below:

- (a) 1-2-3-4 (b) 4-3-2-1
(c) 2-1-3-4 (d) 4-3-1-2

96. Consider the following statements:

1. Neanderthal is a member of a group of archaic humans who emerged during the Pleistocene Epoch.
2. Genome is defined as an organism's complete set of Deoxyribose Nucleic Acid (DNA), including all of its genes.

Which of the statements given above is/are correct?

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

97. Which of the statements given below about Sunderlal Bahuguna is/are correct?

1. He initiated the Chipko Movement to guard trees.
2. He was awarded the Padma Vibhushan.

3. 'Ecology is a permanent economy' slogan was coined by him.

Select the correct answer using the code given below:

- (a) 1 and 2 only (b) 1 only
(c) 1, 2 and 3 only (d) 1 and 3 only

98. Which of the following rivers are the tributaries of the Indus River?

1. Gasting 2. Hunza
3. Zaskar 4. Ravi

Select the correct answer using the codes given below:

- (a) 1, 3 and 4 only (b) 1, 2 and 3 only
(c) 2 and 3 only (d) 1, 2 and 4 only

99. At which one of the following places two important rivers of India originate; while one of them flows towards north and merges with another important river flowing towards Bay of Bengal, the other one flows towards Arabian Sea?

- (a) Amarkantak
(b) Raipur
(c) Mahabaleshwar
(d) Nasik

100. Consider the following statements regarding System of Air Quality and Weather Forecasting And Research (SAFAR):

1. It is an initiative of the Ministry of Environment, Forest and Climate Change.
2. SAFAR is operationalised by the Central Pollution Control Board.
3. The SAFAR system does not monitor the ozone and mercury pollutants.

Which of the statements given above is/are **not** correct?

- (a) 1 and 2 only
(b) 2 only
(c) 3 only
(d) 1, 2 and 3

**Test Code:
2205**

(English Version)

**ANSWER & EXPLANATION
GENERAL STUDIES**

TEST – 05

A

ANSWERS

Q. No.	Answers	Q. No.	Answers	Q. No.	Answers	Q. No.	Answers
1	(d)	26	(b)	51	(d)	76	(c)
2	(a)	27	(a)	52	(a)	77	(b)
3	(b)	28	(a)	53	(d)	78	(c)
4	(c)	29	(c)	54	(b)	79	(c)
5	(a)	30	(d)	55	(c)	80	(c)
6	(d)	31	(a)	56	(b)	81	(d)
7	(b)	32	(a)	57	(a)	82	(a)
8	(b)	33	(c)	58	(a)	83	(c)
9	(d)	34	(b)	59	(b)	84	(d)
10	(c)	35	(a)	60	(c)	85	(c)
11	(d)	36	(a)	61	(c)	86	(b)
12	(d)	37	(d)	62	(a)	87	(d)
13	(b)	38	(b)	63	(c)	88	(c)
14	(c)	39	(c)	64	(d)	89	(a)
15	(a)	40	(a)	65	(b)	90	(b)
16	(b)	41	(a)	66	(c)	91	(c)
17	(c)	42	(d)	67	(a)	92	(c)
18	(d)	43	(d)	68	(b)	93	(b)
19	(b)	44	(b)	69	(c)	94	(b)
20	(a)	45	(b)	70	(c)	95	(d)
21	(d)	46	(b)	71	(c)	96	(c)
22	(b)	47	(d)	72	(b)	97	(c)
23	(c)	48	(a)	73	(a)	98	(b)
24	(c)	49	(c)	74	(a)	99	(a)
25	(c)	50	(b)	75	(c)	100	(d)

1. (d)

Explanation:

- Largest reserves/resources of copper ore to a tune of 813 million tonnes (**53.81%**) are in the state of **Rajasthan** followed by Jharkhand with 295 million tonnes (19.54%) and Madhya Pradesh with 283 million tonnes (18.75%). Copper reserves/ resources in Andhra Pradesh, Gujarat, Haryana, Karnataka, Maharashtra, Meghalaya, Nagaland, Odisha, Sikkim, Tamil Nadu, Telangana, Uttarakhand and West Bengal accounted for remaining 7.9% of the total all India resources.
- Most of the copper reserves are found along the **Aravali range** spread over the districts of Ajmer, Alwar, Bhilwara, Chittorgarh, Dungarpur, Jaipur, Jhunjhunu, Pali, Sikar, Sirohi and Udaipur in Rajasthan. **Hence, option (d) is correct.**

2. (a)

Explanation:

- As per Operational Guidelines, 2019 of UNESCO, it is mandatory to put any monument/site on the Tentative List (TL) for one year before it is considered for the final nomination dossier. **Hence, statement 2 is NOT correct.**
- Once the nomination is done, it is sent to the World Heritage Centre (WHC).
- The World Heritage Centre is the Secretariat to the 1972 Convention. **Hence, statement 1 is correct.**
- **About Six New Places in TL:**
 - Satpura Tiger Reserve (Madhya Pradesh)
 - Ghats of Varanasi (Uttar Pradesh). **Hence, statement 3 is correct.**
 - Megalithic Site of Hirebenkal (Karnataka)
 - Maratha Military Architecture in Maharashtra
 - Bhedaghat-Lametaghat in Narmada Valley- Jabalpur (Madhya Pradesh)
 - Temples of Kanchipuram (Tamil Nadu)

3. (b)

Explanation:

Laterite Soil

- Laterite Soil Laterite has been derived from the Latin word 'Later' which means brick.
- The laterite soils develop in areas with **high temperature and high rainfall.**
- These are the result of intense leaching due to tropical rains.

- With rain, lime and silica are leached away, and soils rich in iron oxide and aluminium compounds are left behind.
- Humus content of the soil is removed fast by bacteria that thrives well in high temperature.
- These soils are poor in organic matter, nitrogen, phosphate and calcium, while iron oxide and potash are in excess.
- Hence, laterites are not suitable for cultivation; however, **application of manures and fertilisers are required for making the soils fertile for cultivation.**
- Red laterite soils in Tamil Nadu, Andhra Pradesh and Kerala are more suitable for tree crops like cashew nut.
- Laterite soils are widely cut as bricks for use in house construction.
- These soils have mainly developed in the higher areas of the Peninsular plateau.
- The laterite soils are commonly found in Karnataka, Kerala, Tamil Nadu, Madhya Pradesh and the hilly areas of Odisha and Assam. **Hence, option (b) is correct.**

4. (c)

Explanation:

- **Gurudev Rabindranath Tagore is said to have composed over 2000 songs and his songs and music are called 'Rabindra Sangeet' with its own distinct lyrical and fluid style.**
- **He is responsible for modernising Bengali prose and poetry. His notable works include Gitanjali, Ghare-Baire, Gora, Manasi, Balaka, Sonar Tori.**
- **He is also remembered for his song 'Ekla Chalo Re'.**
 - He published his first poems aged 16 under the pen name 'Bhanusimha'.
 - He not only gave the national anthems for two countries, India and Bangladesh but also inspired a Ceylonese student of his, to pen and compose the national anthem of Sri Lanka.
 - Besides all his literary achievements he was also a philosopher and educationist who in 1921 established the Vishwa-Bharati University, a university that challenged conventional education. **Hence, statement 1 is correct.**
- **In 1913 he was awarded the Nobel Prize in Literature for his work on Gitanjali.**
 - He was the first non-European to receive the Nobel Prize. **Hence, statement 2 is correct.**
- **In 1915 he was awarded a knighthood by the British King George V. In 1919, following the Jallianwalla Bagh massacre he renounced his Knighthood. Hence, statement 3 is NOT correct.**



5. (a)

Explanation:

Special Window for Affordable & Mid-Income Housing (SWAMIH) Investment Fund:

- This is a government-backed fund that was set up as a Category-II AIF (Alternate Investment Fund) debt fund registered with SEBI, launched in 2019.
- SWAMIH Investment Fund was formed to complete construction of stalled, Real Estate (Regulation and Development) Act (RERA) registered affordable and mid-income category housing projects which are stuck due to paucity of funds. **Hence, option (a) is correct.**
- The Sponsor of the Fund is the Secretary, Department of Economic Affairs, Ministry of Finance, on behalf of the Government of India.
- The residential project - Rivali Park, located in suburban Mumbai, was the first housing project in India to have received funding under the SWAMIH Fund.

6. (d)

Explanation:

Chhattisgarh is bounded by southern **Jharkhand** and **Odisha** in the east, **Madhya Pradesh** and **Maharashtra** in the west, **Uttar Pradesh** and western **Jharkhand** in the north and **Andhra Pradesh** in the south. **Hence, statement 1 is NOT correct.**



Kohima is located to the north of Imphal. **Hence, statement 2 is NOT correct.**

7. (b)

Explanation:

Bugyal: Beyond a certain height in the mountains trees do not grow. The land beyond the treeline is called the “Alpine zone” and the soft green grass or cover on the land is called meadows or Alpine meadows.

- **“Bugyal” is the local name given to Alpine meadows** in Uttarakhand. The word “Bugyal” is derived from the word “bug” or “bugi” which are the local name of the kind of grass which grows in the high mountains.
- Out of the total geographical area of Uttarakhand, **24.11%** consists of **“Alpine Zone”**. But the entire Alpine Zone is not “Bugyal”. This is so because **many parts of the Alpine Zone are permanently under snow cover**, and some are rocky, broken and unstable slopes, which do not qualify as quality meadows. **Hence, statement 1 is NOT correct.**



- A large number of rare and endangered fauna e.g. snow leopard (*Uncia uncia*), blue sheep (*Pseudois nayaur*), Himalayan tahr (*Hemitragus jemlahicus*) among others inhabit the Bugyals during one season or the other.
- Besides, Bugyals are also regarded as repositories of a large number of medicinal and aromatic plants (MAPs) which are used in local medicine as well as for sale to the herbal industries. **Hence, statement 2 is correct.**

8. (b)

Explanation:

Competition Commission of India (CCI)

● About:

○ Statutory Body:

- Competition Commission of India is a statutory body responsible for enforcing the objectives of the Competition Act, 2002.
- CCI was established by the Central Government with effect from 14th October 2003, but it became fully functional on 20th May 2009.
- It has been established under the Ministry of Corporate Affairs. **Hence, statement 1 is NOT correct.**

○ Composition:

- A Chairperson and 6 Members appointed by the Central Government.

● Formation of CCI:

○ The CCI was established under the provisions of the Competition act 2002:

- The Competition Act, 2007, was enacted after amending the Competition Act, 2002, which led to the establishment of the CCI and the Competition Appellate Tribunal.
- The government replaced the Competition Appellate Tribunal (COMPAT) with the National Company Law Appellate Tribunal (NCLAT) in 2017.

● Functions and Role of CCI:

- To eliminate practices having adverse effects on competition, promote and sustain competition, protect the interests of consumers and ensure freedom of trade in the markets of India. **Hence, statement 2 is correct.**
- The Competition Commission of India takes the following measures to achieve its objectives:
 - Consumer welfare to make the markets work for the benefit and welfare of consumers.

- Ensure fair and healthy competition in economic activities in the country for faster and inclusive growth and development of the economy.
- Implement competition policies with an aim to effectuate the most efficient utilization of economic resources.
- Develop and nurture effective relations and interactions with sectoral regulators to ensure smooth alignment of sectoral regulatory laws in tandem with the competition law.
- Effectively carry out competition advocacy and spread the information on benefits of competition among all stakeholders to establish and nurture competition culture in the Indian economy.

9. (d)

Explanation:

- Mizoram which is also known as the 'Molasses basin' which is made up of soft unconsolidated deposits. **Hence, statement 1 is NOT correct.**
- Most of the rivers in Nagaland form the tributary of the Brahmaputra.
- The Barak is an important river in Manipur and Mizoram.
- **Barak River:** The Barak river originates from **Naga Hills**. **It is not a tributary** of brahmaputra. It flows through India and then Bangladesh and falls into the Bay of Bengal. **Hence, statement 2 is NOT correct.**

10. (c)

Explanation:

- Recently, NASA's OSIRIS-REx spacecraft departed from asteroid Bennu and started its two-year-long journey back to Earth.
- The OSIRIS-REx is NASA's first mission to visit a near-Earth asteroid. It is the United States' first **asteroid sample return mission**, aiming to collect and carry a pristine, unaltered sample from an asteroid back to earth for scientific study. **Hence, statement 1 is correct.**
- The OSIRIS-REx (Origins, Spectral Interpretation, Resource Identification, Security, Regolith Explorer) spacecraft was launched in **2016** for the journey to Bennu. **Hence, statement 2 is correct.**
 - The mission is essentially a seven-year-long voyage and will conclude when at least 60 grams of samples are delivered back to the Earth (in 2023).



- As per the National Aeronautics and Space Administration (NASA), the mission promises to bring the largest amount of extraterrestrial material back to the Earth since the Apollo era.
- The spacecraft contains five instruments meant to explore Benu including cameras, a spectrometer and a laser altimeter.
- Recently, the spacecraft's robotic arm called the Touch-And-Go Sample Acquisition Mechanism (TAGSAM), made an attempt to "TAG" the asteroid at a sample site and collected a sample.

11. (d)

Explanation:

- **Tropical Dry Deciduous Forests:** These are similar to moist deciduous forests and shed their leaves in the dry season. The tropical dry deciduous forests are widely distributed over a large area in India.
- Tropical dry deciduous occur in an irregular wide strip running north-south from the foot of the Himalayas to Kanyakumari except in Rajasthan, Western Ghats and West Bengal.
- Area under tropical moist deciduous forests is the largest in India. **Hence, statement 1 is NOT correct.**
- **Littoral and swamps Forests:** These forests occur in and around the **deltas, estuaries and creeks** prone to tidal influences and as such is also known as delta or tidal forests. **Hence, statement 2 is NOT correct.**

12. (d)

Explanation:

Famous Local Storms of Hot Weather Season

- **Mango Shower:** Towards the end of summer, there are pre-monsoon showers which are a common phenomenon in **Kerala and coastal areas of Karnataka**. Locally, they are known as mango showers since they help in the early ripening of mangoes. **Hence, statement 2 is correct.**
Blossom Shower: With this shower, coffee flowers blossom in Kerala and nearby areas.
- **Nor Westers:** These are dreaded evening thunderstorms in **Bengal and Assam**. Their notorious nature can be understood from the local nomenclature of 'Kalbaisakhi', a calamity of the month of Baisakh. These showers are useful for tea, jute and rice cultivation. In **Assam**, these storms are known as "Bardoli Chheerha". **Hence, statement 1 and 3 are correct.**

13. (b)

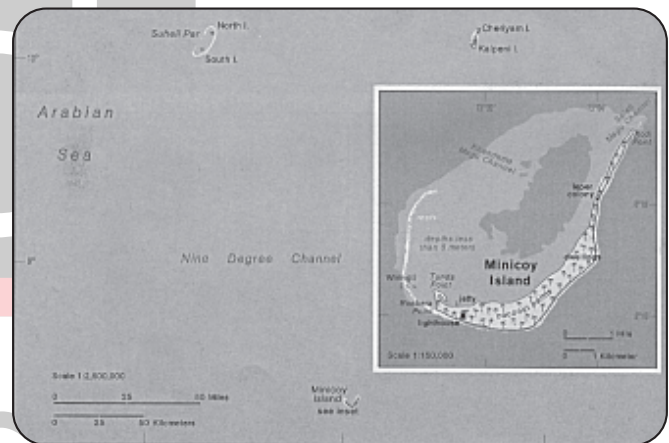
Explanation:

- National Financial Reporting Authority (NFRA) was constituted in 2018 by the Government of India under section 132 (1) of the Companies Act, 2013. It is an audit regulator. **Hence, statement 1 is correct.**
- The NFRA account is monitored by the Comptroller and Auditor General of India. **Hence, statement 2 is correct.**
- NFRA can undertake investigations related to the companies whose securities are listed on any stock exchange in India or outside India. **Hence, statement 3 is NOT correct.**

14. (c)

Explanation:

The Nine Degree Channel separates Minicoy from Kalpeni.



Hence, option (c) is correct.

15. (a)

Explanation:

- Coal is the main source of energy in India as it fulfils almost 67 per cent of the total commercial energy needs of the country. This fossil fuel is found in a form of sedimentary rocks and is often known as 'Black Gold'.
- **Types of coal on the basis of carbon content:**
 - **Anthracite** is the best quality of coal which carries 80 to 95 per cent carbon content.
 - **Bituminous** carries 60 to 80 per cent of carbon content and a low level of moisture content.
 - **Lignite** is often brown in colour. It carries 40 to 55 per cent carbon content.
 - **Peat** has less than 40 per cent of carbon content.



- **Jharkhand** has the **highest** coal reserves in India.
- **Odisha** has the **second** largest coal reserves in the country and it has more than 24 per cent of the total coal reserves.
- **Chhattisgarh** has the **third** largest coal reserve in India and carries about 17 per cent of the total coal reserves. However, the state has the first rank in the production of coal.
- **West Bengal** carries about 11 per cent of the total coal reserves of India. **Hence, option (a) is correct.**

16. (b)

Explanation:

Genetic Modified crops have been engineered in the laboratory in order to favour the expression of desired physiological traits or the production of desired biological products. Examples in food crops include resistance to certain pests, diseases, or environmental conditions, reduction of spoilage, or resistance to chemical treatments (e. g. resistance to herbicide), or improving the nutrient profile of the crop. Examples in non-food crops include production of pharmaceutical agents, bio-fuels, and other industrially useful goods, as well as for bioremediation.

1. **BT cotton:** It is a genetically modified cotton variety, which produces BT toxin, which is insecticidal to **bollworms**. Here, BT refers to *Bacillus thuringiensis*.
2. **Mustard DMH-11:** It is a genetically modified variety of mustard, a **herbicide tolerant crop**, developed by the Centre for Genetic Manipulation of Crop Plants at Delhi University.
3. **Golden rice:** It is a variety of rice (*Oryza sativa*) produced through genetic engineering to biosynthesize beta-carotene, a precursor of **vitamin A**, in the edible parts of rice.
4. **Flavr Savr tomato:** It is a genetically modified food (tomato), engineered to have a longer **shelf life**.

Hence, option (b) is correct.

17. (c)

Explanation:

- Saline soils are also known as **Usara soils**. Saline soils contain a larger proportion of sodium, potassium and magnesium, and thus, they are infertile, and do not support any vegetative growth.

- In the areas of intensive cultivation with **excessive use of irrigation**, especially in areas of green revolution, the fertile alluvial soils are becoming saline.
- Excessive irrigation with dry climatic conditions promotes **capillary action**, which results in the deposition of salt on the top layer of the soil. **Hence, statement 1 is correct.**
- The salt lodged in the lower profiles of the soil comes up to the surface and destroys its fertility. Chemical fertilisers in the absence of organic manures are also harmful to the soil. Unless the soil gets enough humus, chemicals harden it and reduce its fertility in the long run.
- In such areas, especially in Punjab and Haryana, farmers are advised to **add gypsum to solve the problem** of salinity in the soil. **Hence, statement 2 is correct.**

18. (d)

Explanation:

- **Article 311 (1)** says that no government employee either of an all India service or a state government shall be dismissed or removed by an authority subordinate to that appointed him/her.
- **Article 311 (2)** says that no civil servant shall be dismissed or removed or reduced in rank except after an inquiry in which s/he has been informed of the charges and given a reasonable opportunity of being heard in respect of those charges.

○ People Protected under Article 311: The members of

- Civil service of the Union,
- All India Service, and
- Civil service of any State,
- People who hold a civil post under the Union or any State.
- The protective safeguards given under Article 311 are applicable only to civil servants, i.e. public officers. They are not available to defence personnel. **Hence, statement 1 is NOT correct.**
- Article 309 empowers the Parliament and the State legislature to regulate the recruitment, and conditions of service of persons appointed, to public services and posts in connection with the affairs of the Union or of any State respectively.
- According to **Article 310**, except for the provisions provided by the Constitution, a civil servant of the Union works at the pleasure of the President and a civil servant



under a State works at the pleasure of the Governor of that State (English doctrine of Pleasure). **Hence, statement 2 is NOT correct.**

○ But this power of the Government is not absolute.

- Article 311 puts certain restrictions on the absolute power of the President or Governor for dismissal, removal or reduction in rank of an officer.

19. (b)

Explanation:

- Montane ecosystems refers to any ecosystem found in mountains.
- Mountain forests can be classified into two types in India, the **northern** mountain forests and the **southern** mountain forests.
- The Himalayan ranges show a succession of vegetation from the **tropical to the tundra**, which change with the **altitude**.
- Deciduous forests are found in the foothills of the Himalayas. It is succeeded by the wet temperate type of forests between an altitude of 1,000-2,000 m.
- In the higher hill ranges of northeastern India, hilly areas of West Bengal and Uttaranchal, evergreen broad leaf trees such as oak and chestnut are predominant.
- Between 1,500-1,750 m, pine forests are well-developed, with Chir pine being a very useful commercial tree. Deodar, a highly valued endemic species grows mainly in the western part of the Himalayan range. Other important species of Montane forests are blue pine, spruce, silver firs, junipers, pines, birch and rhododendrons. **Hence, statement 2 is correct.**
- The southern slopes of the Himalayas carry a thicker vegetation cover because of relatively higher precipitation than the drier north-facing slopes. At higher altitudes, mosses and lichens form part of the tundra vegetation. **Hence, statements 3 and 4 are correct.**

Southern Mountain Forest

- The southern mountain forests include the forests found in three distinct areas of Peninsular India viz; the Western Ghats, the Vindhyas and the Nilgiris. As they are closer to the tropics, and only 1,500 m above the sea level, vegetation is temperate in the higher regions, and subtropical on the lower regions of the Western Ghats, especially in Kerala, Tamil Nadu and Karnataka. **The temperate forests are called Sholas in the Nilgiris, Anaimalai and Palani hills. Hence, statement 1 is NOT correct.**

20. (a)

Explanation:

- There are two major island groups in India – one in the Bay of Bengal and the other in the Arabian Sea. The Bay of Bengal island groups consist of about 572 islands/islets. These are situated roughly between 6°N-14°N and 92°E -94°E. Bay of Bengal group consists of two principal groups of islets include the **Ritchie's archipelago** and the **Labrynth island**. In this group, the Andaman Islands lie in the north and the Nicobar in the south. **Hence, option (a) is correct.**
- Katchatheevu is an uninhabited islet in the Palk Strait. It is the centre of a long-standing dispute between the fishermen of Tamil Nadu and Sri Lanka.

21. (d)

Explanation:

- 'Prisons/persons detained therein' is a State subject under Entry 4 of List II of the Seventh Schedule to the Constitution of India. **Hence, statement (a) is correct.**
 - Entry 4: Prisons, reformatories, Borstal institutions and other institutions of a like nature, and persons detained therein; arrangements with other States for the use of prisons and other institutions.
 - The administration and management of prisons is the responsibility of respective State Governments.
 - However, the **Ministry of Home Affairs** provides regular guidance and advice to States and UTs on various issues concerning prisons and prison inmates. **Hence, statement (b) is correct.**
- Article 39A (and not Article 23) of the Constitution directs the State to ensure that the operation of the legal system promotes justice on a basis of equal opportunity and shall, in particular, provide free legal aid by suitable legislation or schemes or in any other way, to ensure that opportunities for securing justice are not denied to any citizen by reason of economic or other disabilities. **Hence, statement (d) is NOT correct.**
 - The right to free legal aid or free legal service is an essential fundamental right guaranteed by the Constitution.
 - Article 23 of the Constitution provides the fundamental right against exploitation.



- Article 22 (3) (b) of the Constitution allows for preventive detention and restriction on personal liberty for reasons of state security and public order. **Hence, statement (c) is correct.**
 - Further, Article 22 (4) states that no law providing for preventive detention shall authorise the detention of a person for a longer period than three months unless:

22. (b)

Explanation:

The correct North-to-South Order is:

- Mishmi Hills (2)
- Khasi Hills (4)
- Patkai Bum Hills (1)
- Mizo Hills (3)



Hence, option (b) is correct.

23. (c)

Explanation:

Pradhan Mantri Awaas Yojana-Gramin:

- It was launched to achieve the objective of “Housing for All” by 2022, the erstwhile rural housing scheme Indira Awaas Yojana (IAY) was restructured to Pradhan Mantri Awaas Yojana-Gramin (PMAY-G) w.e.f 1st April 2016.
- It comes under the Ministry of Rural Development. **Hence, statement 3 is correct.**
- It aims to provide a pucca house with basic amenities to all rural families, who are homeless or living in kutcha or dilapidated houses by the end of March 2022. **Hence, statement 2 is NOT correct.**

Beneficiaries:

- Beneficiaries under PMAY (G) will include all the houseless and households living in zero, one or two-room houses with kutcha wall and kutcha roof as per SECC-2011 data. **Hence, statement 1 is NOT correct.**
- Priority will first be assigned on the basis of parameters reflecting housing deprivation in each category viz., SC/ST, Minorities and others.

24. (c)

Explanation:

- A stretch of **Dihing or Burhi Dihing** in Eastern Assam’s Dibrugarh district witnessed fire following the spillage of crude oil which was being transported to Digboi refinery.

Dihing River:

- Dihing or Burhi Dihing is a **large tributary of the Brahmaputra River** in Upper Assam. **Hence, statement 1 is correct.**
- The river originates in the **Eastern Himalayas (the Patkai Hills)** in Arunachal Pradesh and flows through Tinsukia and Dibrugarh Districts in Assam to its confluence with the Brahmaputra at Dihingmukh. **Hence, statement 2 is correct.**
- It creates a number of **oxbow lakes** in the area.



25. (c)

Explanation:

- The Peninsular Plateau constitutes one of the **ancient landmasses** on the earth’s surface.



- It is supposed to be one of the **most stable land blocks on Earth**. Hence, **statement 2 is NOT correct**.
- The Peninsular plateau is a tableland composed of the **old crystalline, igneous and metamorphic rocks**. Hence, **statement 1 is correct**.
- The plateau has **broad and shallow valleys and rounded hills**. Hence, **statement 3 is correct**.
- It was formed due to the **breaking and drifting of the Gondwana land** and thus, making it a part of the oldest landmass.
- This plateau consists of **two broad divisions**, namely, the **Central Highlands** and the **Deccan Plateau**.
- The part of the Peninsular plateau lying to the **north of the Narmada river**, covering a major area of the **Malwa plateau**, is known as the **Central Highlands**.

26. (b)

Explanation:

- Important minerals used for the generation of nuclear energy are uranium and thorium. Uranium deposits occur in the Dharwar rocks.
- Geographically, uranium ores are known to occur in several locations along the Singbhum Copper belt.
- It is also found in Udaipur, Alwar and Jhunjhunu districts of Rajasthan, Durg district of Chhattisgarh, Bhandara district of Maharashtra and Kullu district of Himachal Pradesh. Hence, **statement 1 is NOT correct**.
- Uranium Corporation of India Limited is a Public Sector Enterprise under the Department of Atomic Energy with a special standing at the forefront of Nuclear Power cycle.
- The Company operates six underground mines (Bagjata, Jaduguda, Bhatin, Narwapahar, Turamdih and Mohuldih) and one open pit mine (Banduhurang) in the State of Jharkhand.
- Thorium is mainly obtained from monazite and ilmenite in the beach sands along the coast of Kerala and Tamil Nadu. World's richest monazite deposits occur in Palakkad and Kollam districts of Kerala, near Vishakhapatnam in Andhra Pradesh and Mahanadi river delta in Odisha. Hence, **statement 2 is correct**.
- Indian Rare Earths Limited (IREL) is engaged in mining and separation of beach sand minerals in the southern part of the country.

27. (a)

Explanation:

- Cyclone Tauktae was a tropical cyclone, named by **Myanmar**. It means 'gecko', a highly vocal lizard, in the Burmese language. Hence, **statement 2 is NOT correct**.
- Typically, tropical cyclones in the North Indian Ocean region (Bay of Bengal and Arabian Sea) develop during the pre-monsoon (April to June) and post-monsoon (October to December) periods.
- A tropical cyclone is an intense circular storm that originates over warm tropical oceans and is characterized by low atmospheric pressure, high winds, and heavy rain. Hence, **statement 1 is correct**.
- A characteristic feature of tropical cyclones in the eye, a central region of clear skies, warm temperatures, and low atmospheric pressure.
- According to WMO (World Meteorological Organization) guidelines, countries in every region are supposed to give names for cyclones.
- The North Indian Ocean Region covers tropical cyclones formed over the Bay of Bengal and the Arabian Sea.
- The 13 members, which come under the region, are Bangladesh, India, Maldives, Myanmar, Oman, Pakistan, Sri Lanka, Thailand, Iran, Qatar, Saudi Arabia, the UAE and Yemen.
- The India Meteorological Department (IMD), one of the six Regional Specialised Meteorological Centres (RSMC) in the world, is mandated to issue advisories and name tropical cyclones in the North Indian Ocean Region. Hence, **statement 3 is correct**.

28. (a)

Explanation:

- The **Narmada** rises in the **Amarkantak hills in Madhya Pradesh**. It flows towards the west in a rift valley formed due to faulting. On its way to the sea, the Narmada creates many picturesque locations. Hence, **pair 1 is correctly matched**.
- The **Godavari** is the **largest Peninsular river**. It rises from the slopes of the Western Ghats in **Nasik district of Maharashtra**. Its length is about 1500 km. It drains into the Bay of Bengal. Its drainage basin is also the largest among the peninsular rivers.



- Rising from a spring near **Mahabaleshwar**, the **Krishna** flows for about 1400 km and reaches the Bay of Bengal. **Hence, pair 2 is NOT correctly matched.**
- The **Kaveri** rises in the **Brahmagiri range** of the **Western Ghats** and it reaches the Bay of Bengal in the south of Cuddalore in Tamil Nadu. The total length of the river is about 760 km. **Hence, pair 3 is NOT correctly matched.**



29. (c)

Explanation:

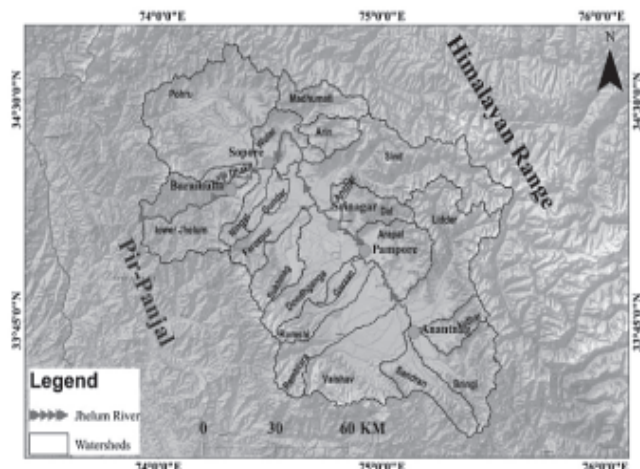
Jhelum and Meanders

Jhelum in the valley of Kashmir, still in its youth stage, forms meanders – a typical feature associated with the mature stage in the evolution of fluvial landform. **Hence, statement 1 is correct.**

In Kashmir Valley, the meanders in Jhelum river are caused by the local base level provided by the erstwhile larger lake of which the present Dal Lake is a small part. **Hence, statement 2 is correct.**

Local Base Level: The local base level can be defined as the level where the velocity and eroding power of the water is temporarily lost.

Examples of local base levels include a lake, a dam, and a waterfall. When streams flow into these areas, the stream has reached the lowest level at which it can flow.



30. (d)

Explanation:

- Ferrous Metals mostly contain Iron. They have small amounts of other metals or elements added, to give the required properties. Ex: Iron ore, Manganese, Chromite, etc. **Hence, statement 1 is correct.**
- As per United Nations Framework classification (UNFC) of mineral resources, total resources of iron ore in the country is around 28.52 billion tonnes as on 1.4.2010.
- The two main types of iron ore found in our country are **haematite** (metallic content varies between 60-70%) and **magnetite** (metallic content varies between 60-65%). About 17.88 billion tonnes of haematite and about 10.64 billion tonnes of magnetite are available in the country. **Hence, statement 2 is correct.**
- Other types of iron-ore are: Limonite (35-40% of metal) and Siderite (10-40% of metal).
- About 95 per cent of total reserves of iron ore is located in the States of Odisha, Jharkhand, Chhattisgarh, Karnataka, Goa, Telangana, Andhra Pradesh and Tamil Nadu.
- In Odisha, iron ore occurs in a series of hill ranges in Sundergarh, Mayurbhanj and Jhar. The important mines are Gurumahisani, Sulaipet, Badampahar (Mayurbhanj), Kiruburu (Kendujhar) and Bonai (Sundergarh).
- In Chhattisgarh, Bailadila in the Bastar district and Dalli-Rajhara in the Durg district are the main iron-ore producing regions. **Hence, statement 3 is correct.**
- In Jharkhand, Noamundi, Daltenganj (Palamau district) are the famous mines.



- In Karnataka, iron ore deposits occur in Sandur-Hospet area of Ballari district, Baba Budan hills and Kudremukh in Chikkamangaluru district and parts of Shivamogga, Chitradurg and Tumakuru districts.
- The districts of Chandrapur, Bhandara and Ratnagiri in Maharashtra, Karimnagar and Warangal district of Telangana, Kurnool, Cuddapah and Anantapur districts of Andhra Pradesh, Salem and Nilgiris districts of Tamil Nadu are other iron mining regions.

31. (a)

Explanation:

- Cotton is a plant of Tropical and Sub-Tropical regions.
- It requires average temperatures between 21°C and 30°C.
- Maritime air promotes blossoming and improves quality of fibre.
- Frost is detrimental to the crop.
- It requires 50 to 75 cm of rainfall but not more than 85 cm.
- When the climate is humid, it is suitable for the production of cotton. **Hence, statement 1 is correct.**
- Cultivation of Cotton is mainly concentrated in the area where black soil is found. It is the most suitable soil type that supports the cultivation of Cotton. But the crop can also grow in alluvial soil. **Hence, statement 2 is correct.**
- Gujarat is the leading state in the production of cotton followed by Maharashtra. **Hence, statement 3 is NOT correct.**

32. (a)

Explanation:

- In 1973, the Dhebar Commission created Primitive Tribal Groups (PTGs) as a separate category, which are less developed among the tribal groups.
- In 2006, the Government of India renamed the PTGs as PVTGs. **Hence, statement 1 is correct.**
 - In 1975, the Government of India initiated to identify the most vulnerable tribal groups as a separate category called PVTGs and declared 52 such groups. In 1993 an additional 23 groups were added to the category, making it a total of 75 PVTGs out of 705 Scheduled Tribes.
 - Among the 75 listed PVTG's the highest number are found in Odisha.
 - The 13 PVTGs in Odisha are: Bonda, Birhor, Chuktia Bhunjia, Didayi, Dungaria Kandha, Hill Kharia, Juang, Kutia Kondh, Lanjia Saora, Lodha, Mankirdia, Paudi Bhuyan and Saora.

- The PVTGs in Andaman & Nicobar Islands are: Great Andamanese, Jarawas, Onges, Sentinelese and Shorn Pens. **Hence, statement 2 is NOT correct.**

33. (c)

Explanation:

The Veliconda Range or Veliconda Hills is a low mountain range that forms part of the **Eastern Ghats mountain range system**. The Veliconda Range is located in the south-eastern part of Andhra Pradesh state. Peninsular India mostly consists of relict and residual mountains like the Nallamala hills, the Javadi hills, the Veliconda hills, the Palkonda range and the Mahendragiri hills, etc. **Hence, option (c) is correct.**

34. (b)

Explanation:

- The Banni grassland in the **Kachchh District of Gujarat** is an area of 2500 square kilometre and is often referred to as the **largest tropical grassland in Asia**. **Hence, both statements 1 and 2 are correct.**
- Banni is home to great biological diversity.
- Banni Grassland having numerous large, medium and small sized wetlands attracts migratory birds to breed and rest in winter.
- The Kachchh Desert Wildlife Sanctuary and Chhari Dhand Conservation Reserve are part of the Banni Grassland.
- The **Maldharis** are the nomadic pastoralist natives of Banni. **Hence, statement 3 is NOT correct.**
- The Maldharis rights to graze the Banni are derived from historical rulers of the region who granted these rights in return for a grazing tax.
- The region also serves as a breeding ground for the Banni buffalo and the Kankrej cow, for sale in many others parts of the country.
- Recently this grassland is in news because it is in the grip of a severe drought as winter sets in.
- While the area is known for its scarce rainfall and semi-drought conditions, it is for the first time in 13 years that the Maldharis, the nomadic natives of Banni, are leaving their villages as water gets scarce.
- The **Rabari, also called the Rewari or Desai**, are an indigenous tribal caste of nomadic cattle and camel herders and shepherds that live throughout northwest India, primarily in the states of **Gujarat, Punjab and Rajasthan**.



35. (a)

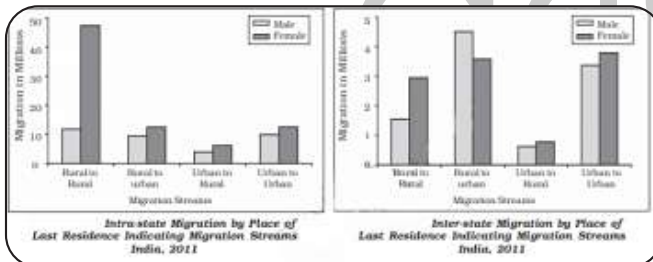
Explanation:

- The United Nations General Assembly (UNGA) in 1993 proclaimed 22nd May as International Day for Biological Diversity (IDB) to increase understanding and awareness of biodiversity issues. **Hence, statement 1 is correct.**
- The UNGA also declared the period 2011-2020 as the United Nations (UN) Decade on Biodiversity to promote the implementation of a strategic plan on biodiversity and its overall vision of living in harmony with nature.
- It also declared 2021-2030 as the UN Decade of Ocean Science for Sustainable Development and the UN Decade on Ecosystem Restoration. **Hence, statement 3 is correct.**
- This year's theme is "We're part of the solution" which is a continuation of the 2020 theme. Our solutions are in nature. **Hence, statement 2 is NOT correct.**
- This serves as a reminder that biodiversity remains the answer to several sustainable development challenges.

36. (a)

Explanation:

- In India during 2011, out of 315 million migrants, enumerated on the basis of the last residence, 98 million had changed their place of residence in the last ten years.
- Out of these, 81 million were intrastate migrants. The stream was dominated by female migrants. Most of these were migrants related to marriage.
- The distribution of male and female migrants in different streams of intra-state and inter-state migration is presented in following figures:



- It is clearly evident that females predominate the streams of short distance rural to rural migration in both types of migration.
- Contrary to this, men predominate the rural to urban stream of inter-state migration due to economic reasons. **Hence, option (a) is correct.**
- **Extra:** According to new Census 2011 data, 45.36 crore Indians (37 per cent) in India are migrants — now settled in a place different from their previous residence.

- Between 2001-2011, marriage was the dominant reason for migration among women, as was the case in previous two decades. Around 21.7 crore of the 31.2 crore female migrants — 69.7 per cent — cited this reason. 65.9 per cent women in 1981-1991 and 64.9 per cent in 1991-2001 migrated due to the same reason.
- For men, 'work and employment' was the top reason, mentioned by three crore of the 14 crore male migrants.

37. (d)

Explanation:

The National Bureau of Soil Survey and the Land Use Planning, an institute under the control of the Indian Council of Agricultural Research (ICAR) has classified the Indian soils on the basis of their nature and character as per the United States Department of Agriculture (USDA) Soil Taxonomy. **Hence, option (d) is correct.**

ICAR has classified the soils of India into the following order as per the USDA soil taxonomy

S. No.	Order	Area (In Thousand Hectares)	Percentage
(i)	Inceptisols	130372.90	39.74
(ii)	Entisols	92131.71	28.08
(iii)	Alfisols	44448.68	13.55
(iv)	Vertisols	27960.00	8.52
(v)	Aridisols	14069.00	4.28
(vi)	Ultisols	8250.00	2.51
(vii)	Mollisols	1320.00	0.40
(viii)	Others	9503.10	2.92
Total		100	

Source: Soils of India, National Bureau of Soil Survey and Land Uses Planning, Publication Number 94.

38. (b)

Explanation:

- There is a process as per the Juvenile Justice (JJ) Act, 2015 which needs to be followed with children who have been orphaned. **Hence, statement 1 is correct.**
 - If someone has information about a child in need of care, then they must contact one of the four agencies: Childline 1098, or the district Child Welfare



Committee (CWC), District Child Protection Officer (DCPO) or the helpline of the State Commission for Protection of Child Rights.

- Following this, the CWC will assess the child and place him or her in the immediate care of a Specialised Adoption Agency.
- The State thus takes care of all such children who are in need of care and protection, till they turn 18 years.
- Once a child is declared legally free for adoption by the CWC, adoption can be done either by Indian prospective adoptive parents or non-resident Indians or foreigners, in that order.
 - India has ratified Hague Convention on Intercountry Adoptions, 1993. **Hence, statement 2 is NOT correct.**
- The Central Adoption Resource Authority (CARA), a statutory body of the Ministry of Women and Child Development, is the nodal agency for adoption (and not the State Commission for Protection of Child Rights). **Hence, statement 3 is NOT correct.**
 - It regulates the adoption of orphaned, abandoned and surrendered children through its associated or recognised agencies.

39. (c)

Explanation:

From north to south the hills are:

Nilgiri, Palani, Annamalai and Cardamom. Palghat pass between Nilgiri and Anamalai hills. Nallamala, Javadi too are hill ranges in the south but above the above hills.

The western and the eastern ghats meet at Nilgiri hills.

Hence, option (c) is correct.

40. (a)

Explanation:

A **river delta** is a landform that forms at the mouth of a river, where the river flows into an ocean, sea, estuary, lake, or reservoir. Deltas form from deposition of sediment carried by a river as the flow leaves its mouth. Over long periods, this deposition builds the characteristic geographic pattern of a river delta.

The following are conditions necessary for formation of river delta:

- Active vertical and lateral erosion in the upper course of the river to provide extensive sediments to be eventually deposited as deltas.
- There should not be extensive deposition in the middle stage e.g. presence of lake in between or high evaporation rate (first).
- The river's load must be deposited faster than it can be removed by the action of currents and tides.
- The velocity of a river must be sufficiently low to allow most of its load to be deposited in the river's mouth. **Hence, statement 2 is correct.**
- The coast should be preferably tide less. **Hence, statement 3 is NOT correct.**
- The sea adjoining the delta should be shallow or else the load will disappear in the deep waters. **Hence, statement 1 is correct.**
- There should be no large lakes in the river course to filter-off the sediments and there should be no strong current running at right angles to the river mouth, washing away the sediments.

41. (a)

Explanation:

CoviSelf:

- It is developed by MyLab Discovery Solutions, a Pune-based company.
- It is India's first self-use Rapid Antigen Test (RAT). **Hence, option (a) is correct.**
- This test is synced with a mobile app, CoviSelf, which will help directly feed the positive case's report on the ICMR portal.
- ICMR has advised this test only for those who have symptoms or are high-risk contacts of positive patients and need to conduct a test at home.

42. (d)

Explanation:

Type of Forest	Endemic Trees
Montane	Silver Fir, Junipers , Pine and Birch
Tropical Deciduous	Teak, Bamboo, Sal, Shisham , Sandalwood, Khair, Mulberry
Tropical Evergreen	Mahogany , Ebony, Rosewood, Rubber

Hence, option (d) is correct.



43. (d)

Explanation:

The Indian States bordering Myanmar are:

- Arunachal Pradesh
- Manipur
- Nagaland
- Mizoram



Hence, option (d) is correct.

Tripura shares its border with Bangladesh.

44. (b)

Explanation:

International Museum Day

- International Museum Day is observed on 18th May every year to raise awareness among people about the museums. Hence, statement 1 is correct.
- Theme for 2021: The Future of Museums: Recover and Reimagine.
- The Day was instituted in 1977 by the International Council of Museums (ICOM). Hence, statement 2 is correct.
- Multiple ministries hold charges of various Museums.
- Not all Museums are administered by the Ministry of Culture. Hence, statement 3 is NOT correct.
- Some are run without government support by a handful of people under a Board of Trustees.

45. (b)

Explanation:

Out of the given options only **Subarnarekha** and **Penner** drain into the Bay of Bengal. Hence, option (b) is correct.

Indus, Tapti, Mahi, Narmada and Sabarmati drain in the Arabian Sea.



46. (b)

Explanation:

- It is to be noted that the percentage of area prone to flooding is the highest in **Uttar Pradesh**. Bihar has the second-largest flood-prone area. Hence, statement 1 is NOT correct.
- In West Bengal, the southern and central parts are flooded by the Mahananda, Bhagirathi, Ajoy, Damodar etc. due to inadequate capacity of river channels and **tidal effect**. Hence, statement 2 is correct.

47. (d)

Explanation:

- Under Article 171 of the Constitution, the Legislative Council of a state shall not have more than **one-third** of the total strength of the State Assembly, and not less than 40 members. Hence, statement 1 is correct.
 - As the Rajya Sabha, the legislative council is a continuing chamber, that is, it is a permanent body and is not subject to dissolution. The tenure of a Member of the Legislative Council (MLC) is six years, with one-third of the members retiring every two years.
- **Manner of Election:**
 - One-third of the MLCs are elected by the state's MLAs.



- Another 1/3rd by a special electorate comprising sitting members of local governments such as municipalities and district boards. **Hence, statement 3 is correct.**
- 1/12th by an electorate of teachers and another 1/12th by registered graduates.
- The remaining members are appointed by the Governor for distinguished services in various fields namely, literature, science, art, cooperative movement and social service.
- The legislative power of the Councils are limited. Unlike Rajya Sabha which has substantial powers to shape non-financial legislation, Legislative Councils lack a constitutional mandate to do so.
- Assemblies can override suggestions/amendments made to legislation by the Council.
- Unlike Rajya Sabha MPs, **MLCs cannot vote** in elections for the **President and Vice President**. The Vice President is the Rajya Sabha Chairperson while a member from the Council itself is chosen as the Council Chairperson. **Hence, statement 2 is correct.**

48. (a)

Explanation:

Visakhapatnam Port

- Visakhapatnam Port in Andhra Pradesh is a **land-locked harbour**, connected to the sea by a channel cut through solid rock and sand. **Hence, pair 2 is NOT correctly matched.**
- An outer harbour has been developed for handling iron-ore, petroleum and general cargo.
- Andhra Pradesh and Telangana are the main hinterland for this port.

Kochchi Port

- Kochchi Port, situated at the head of **Vembanad Kayal**, popularly known as the 'Queen of the Arabian Sea', is also a natural harbour. **Hence, pair 3 is NOT correctly matched.**
- This port has an advantageous location being close to the Suez-Colombo route.
- It caters to the needs of Kerala, southern- Karnataka and south western Tamil Nadu.

Marmagao Port

- Marmagao Port, situated at the entrance of the **Zuari estuary**, is a natural harbour in Goa. **Hence, pair 1 is correctly matched.**

- It gained significance after its remodelling in 1961 to handle iron-ore exports to Japan. Construction of Konkan railway has considerably extended the hinterland of this port.
- Karnataka, Goa, Southern Maharashtra constitute its hinterland.



49. (c)

Explanation:

- The **International Council of Museums (ICOM)** is a membership association and a non-governmental organisation which establishes professional and ethical standards for museum activities (and not UNESCO). It is the only global organisation in the museum field. **Hence, statement 1 is NOT correct.**
 - It was created in 1946 and is headquartered in Paris, France.
 - It serves as a network of museum professionals (more than 40,000 members in over 138 countries).
 - The ICOM Red Lists of Cultural Objects at Risk are practical tools to curb the illegal traffic of cultural objects.
 - ◆ Red Lists present the categories of cultural objects that can be subjected to theft and traffic.
- In India, multiple ministries hold charges of various Museums.
 - Not all Museums are administered by the Ministry of Culture.
 - Some are run without government support by a handful of people under a Board of Trustees.



- **Article 49** of the DPSP of Indian Constitution directs the State to protect monuments, places and objects of artistic or historic interest which are declared to be of national importance. **Hence, statement 2 is correct.**
- The Ministry of Culture provides financial assistance under the Museum Grant Scheme to the State Governments and Societies, Autonomous bodies, Local Bodies and Trusts registered under the Societies Act, for setting up new Museums.
- The National Portal and Digital Repository for Indian Museums (under the Ministry of Culture) have been launched for digitisation of the collections of the Museums. **Hence, statement 3 is correct.**

50. (b)

Explanation:

- Majority of glaciers in India are located in the **Karakoram range** it is home to around 16000 square kilometres of glaciers, around half of the snowbound area of the Himalayas. **Hence, statement 1 is NOT correct.**
- Gangotri glacier is located in the Kumaon-Garhwal region of Himalayas. **Hence, statement 2 is correct.**

51. (d)

Explanation:

Black Soil

- Black soil covers most of the Deccan Plateau which includes parts of Maharashtra, Madhya Pradesh, Gujarat, Andhra Pradesh and some parts of Tamil Nadu.
- In the upper reaches of the Godavari and Krishna, and the north western part of the Deccan Plateau, the black soil is very deep.
- These soils are also known as the '**Regur Soil**' or the 'Black Cotton Soil'. **Hence, statement 1 is correct.**
- The black soils are generally **clayey, deep and impermeable**.
- They swell and become **sticky when wet** and shrink when dried. So, during the dry season, these soil develop wide cracks. Thus, there occurs a kind of 'self ploughing'. **Hence, statement 2 is correct.**
- Black soil has a characteristic of slow absorption and loss of moisture, the black soil retains the moisture for a very long time, which helps the crops, especially the rain fed ones, to sustain even during the dry season. **Hence, statement 3 is NOT correct.**

- Chemically, the black soils are rich in lime, iron, magnesia and alumina. They also contain potash.
- But they lack in phosphorus, nitrogen and organic matter. The colour of the soil ranges from deep black to grey.

52. (a)

Explanation:

Project Elephant:

- Project Elephant was launched in **1992** to provide financial and technical support to wildlife management efforts by states for their free-ranging populations of wild Asian Elephants. **Hence, statement 1 is correct.**
- It is a **Centrally Sponsored Scheme (CSS)**.
- **Elephant Census:**
 - The elephant census is conducted **once in 5 years** under the aegis of Project Elephant. The last Elephant Census was conducted in 2017.
 - According to the Elephant Census 2017, India's Asian elephant population is 27,312.
 - The numbers are lower than from the last census estimate in 2012 (between 29,391 and 30,711).
 - Karnataka has the highest number of elephants, followed by Assam and Kerala. **Hence, statement 2 is NOT correct.**
 - **Elephant Reserves:**
 - There are around 32 Elephant Reserves in India. The very first elephant reserve was the Singhbhum Elephant Reserve of Jharkhand.

● Protection Status of Asian Elephants:

- **IUCN Red List:** Endangered. **Hence, statement 3 is NOT correct.**
- CITES Appendix I
- India's Wildlife (Protection) Act, 1972: Schedule I.

● Related Global Initiative:

- The Monitoring the Illegal Killing of Elephants (MIKE) programme, launched in 2003, is an international collaboration that tracks trends in information related to the illegal killing of elephants from across Africa and Asia, to monitor the effectiveness of field conservation efforts.

53. (d)

Explanation:

Tundra

- Northern and Southernmost region of the world adjoining the ice bound poles.



- **Devoid of trees except stunted shrubs** in the southern part of tundra biome, **ground flora includes lichen, mosses and sedges**. The typical animals are reindeer, arctic fox, polar bear, snowy owl, lemming, arctic hare, and ptarmigan. Reptiles and amphibians are almost absent.

Taiga

- The dominating vegetation is coniferous evergreen mostly spruce, with some pine and firs. **Hence, statement 1 is NOT correct.**

Grasslands:

- Grassland biomes consist of large open areas of grass. Trees can be present, but they are infrequent. Low rainfall, wildland fires, and grazing by animals are three factors that maintain grasslands. **Hence, statement 2 is NOT correct.**
- In grassland regions, the climate is ideal for the growth of grasses only. The low precipitation rates are enough to nourish grasses but not enough for a forest of trees. Frequent fires also play a role in maintaining grassland ecosystems.
- Grasses are well adapted to grow back after a fire. Grassland animals are also prepared for fires, fleeing or burrowing underground to wait out the flames.
- Large animals, such as African elephants, can also trample the ground and discourage the growth of trees.

54. (b)

Explanation:

- **Zojila pass** is situated in the Union Territory of Ladakh between Srinagar and Leh.
- **Nathu la Pass** located in **Sikkim** connects India and China and is used for pilgrimages to Mansarovar Lake.
- **Rohtang la Pass** is located in **Himachal Pradesh** it connects Manali to Keylong.
- **Lipulekh Pass** is located in **Uttarakhand**. **Hence, option (b) is correct.**

55. (c)

Explanation:

- Crime and Criminal Tracking Network and Systems (CCTNS) is a centrally funded scheme, developed by the National Crime Records Bureau (NCRB). **Hence, statement 1 is correct.**

- It is a Mission Mode Project under the National e-Governance Plan (NeGP) of the Ministry of Home Affairs. **Hence, statement 2 is NOT correct.**
 - It was approved in 2009.
 - It is a secure application that links over 97% of police stations in the country.
- **Objectives of CCTNS:**
 - CCTNS aims to make the Police functioning citizen friendly and more transparent by automating the functioning of Police Stations. **Hence, statement 3 is correct.**
 - Improve delivery of citizen-centric services through effective usage of ICT.
 - Provide the Investigating Officers of the Civil Police with tools, technology and information to facilitate investigation of crime and detection of criminals.

56. (b)

Explanation:

The Correctly Matched Pairs are:

1. Vishnu Prayag – Alaknanda and Dhaul Ganga
2. **Nand Prayag – Alaknanda and Nandakini**
3. Karna Prayag – Alaknanda and Pindar
4. Rudraprayag – Alaknanda and Mandakini
5. Devprayag – Alaknanda and Bhagirathi



Hence, option (b) is correct.

57. (a)

Explanation:

- Rainforests are Earth's oldest living ecosystems, with some surviving in their present form for at least 70 million years. **Hence, statement 2 is NOT correct.**
- They are incredibly diverse and complex, home to more than half of the world's plant and animal species—even though they cover just 6% of Earth's surface.



- **Rainforests thrive on every continent except Antarctica. Hence, statement 1 is correct.**
- The largest rainforests on Earth surround the Amazon River in South America and the Congo River in Africa.
- The tropical islands of Southeast Asia and parts of Australia support dense rainforest habitats. Even the cool evergreen forests of North America's Pacific Northwest and Northern Europe are a type of rainforest.
- Rainforests' rich biodiversity is incredibly important to our well-being and the well-being of our planet. Rainforests help regulate our climate and provide us with everyday products.

58. (a)

Explanation:

Key Points

- **Di-Ammonium Phosphate (DAP):**
 - DAP is the second most commonly used fertiliser in India after urea. **Hence, statement 1 is correct.**
 - Farmers normally apply this fertiliser just before or at the beginning of sowing, as it is high in phosphorus (P) that stimulates root development.
 - DAP (46% P, 18% Nitrogen) is the preferred source of Phosphorus for farmers. This is similar to urea, which is their preferred nitrogenous fertiliser containing 46% N.
- **About Subsidy Scheme for Fertilisers:**
 - Under the current scheme, the MRP of Urea is fixed but the subsidy can vary while MRP of DAP is decontrolled (i.e subsidy is fixed but the MRP can vary).
 - All Non-Urea based fertilisers are regulated under the Nutrient Based Subsidy Scheme. **Hence, statement 2 is NOT correct.**
- **About Nutrient-Based Subsidy (NBS) Regime:**
 - Under the NBS regime – fertilizers are provided to the farmers at subsidized rates based on the nutrients (N, P, K & S) contained in these fertilizers.
 - Also, the fertilizers which are fortified with secondary and micronutrients such as molybdenum (Mo) and zinc are given additional subsidies.
 - The subsidy on Phosphatic and Potassic (P&K) fertilizers is announced by the Government on an annual basis for each nutrient on a per kg basis – which is determined taking into account the international and domestic prices of P&K fertilizers, exchange rate, inventory level in the country etc.

- NBS policy intends to increase the consumption of P&K fertilizers so that optimum balance (N:P: K= 4:2:1) of NPK fertilization is achieved.
 - This would improve soil health and as a result, the yield from the crops would increase, resulting in enhanced income to the farmers.
 - Also, as the government expects rational use of fertilizers, this would also ease off the burden of fertilizer subsidy.
- It is being implemented from April 2010 by the Department of Fertilizers, Ministry of Chemicals & Fertilizers.

59. (b)

Explanation:

- Harmonised Master List of Infrastructure sub-sectors is notified by the Ministry of Finance. **Hence, statement 1 is NOT correct.**
- **It includes the following categories:**
 - **Transport and Logistics:** Roads and bridges, Inland waterways, Airport, etc.
 - **Energy:** Electricity Generation, Electricity Transmission, etc.
 - **Water and Sanitation:** Solid Waste Management, Water treatment plants, etc.
 - **Communication:** Telecommunication, etc.
 - **Social and Commercial Infrastructure:** Education Institutions (capital stock), Sports Infrastructure, Hospitals (capital stock), Tourism infrastructure, etc. **Hence, statement 2 is correct.**
- Recently, the Finance Ministry has granted 'Infrastructure' status for exhibition and convention centres.
 - It has been included in the list by insertion of a new item in the category of Social and Commercial Infrastructure.
- In 2020, the government added affordable rental housing projects to the list of sectors recognised as infrastructure.

60. (c)

Explanation:

E-Way Bill (EWB):

- EWB is a unique bill number generated for the specific consignment involving the movement of goods.



- Under the Goods and Services Tax (GST) regime, EWBs are mandatory for inter-state transportation of goods valued over Rs. 50,000 from April 2018, with the exemption to precious items such as gold. **Hence, statement 1 is correct.**
- It is a mechanism to ensure that goods being transported comply with the GST Law and is a tool to track the movement of goods and check tax evasion. **Hence, statement 2 is correct.**

61. (c)

Explanation:

Types of Irrigation

- On the basis of the main source of moisture for crops, the farming can be classified as irrigated and rainfed (barani).
- There is a difference in the nature of irrigated farming, as well as based on the **objective of irrigation, i.e., protective or productive.**
- The objective of **protective irrigation** is to protect the crops from **adverse effects of soil moisture** deficiency which often means that irrigation acts as a supplementary source of water over and above the rainfall. **Hence, statement 1 is correct.**
- The strategy of this kind of irrigation is to provide soil moisture to the maximum possible area.
- **Productive irrigation** is meant to **provide sufficient soil moisture** in the cropping season to achieve high productivity.
- In such irrigation the water input per unit area of cultivated land is higher than protective irrigation. **Hence, statement 2 is correct.**

62. (a)

Explanation:

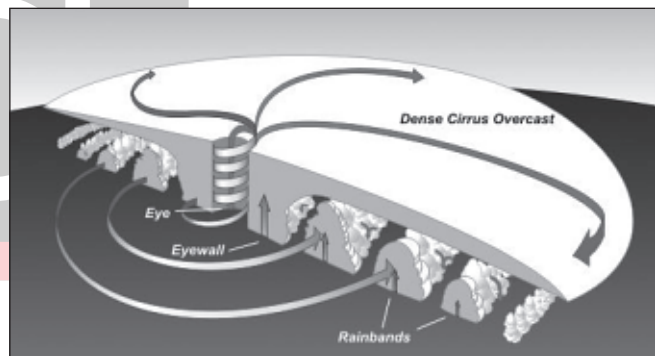
- The Transgender Persons (Protection of Rights) Act, 2019 declared that a transgender person is one whose gender does not match the gender assigned at birth. It includes transmen and trans-women, persons with intersex variations, gender-queers, and persons with socio-cultural identities.
 - In the case National Legal Services Authority (NALSA) v. Union of India, 2014, the Supreme Court declared transgender people to be a 'third gender'. **Hence, statement 1 is NOT correct.**
- The act provides the right of residence with parents and immediate family members.

- It prohibits discrimination against a transgender person in various sectors such as education, employment, and healthcare etc.
- The Act established the National Council for Transgender persons. It functions under the Ministry of Social Justice and Empowerment. **Hence, statement 2 is correct.**

63. (c)

Explanation:

- Tropical Cyclones which are known as Hurricanes in North America and Typhoon in China usually develop in late summers and always emerge from Oceans.
- Due to high Ocean temperature in the summers a low-pressure system develops. As the system intensifies a cumulonimbus cloud is formed and circulating air rises above.
- Tropical Cyclones are more devastating than Temperate Cyclones.



- A fully developed tropical cyclone has a central calm area where the weather is quiet with calm winds. This area is known as the eye of the cyclone with a diameter varying from 10 to 100 km. **Hence, statement 1 is NOT correct.**
- The pressure is lowest and the temperature is highest in the central region with either clear or partly cloudy skies. **Hence, statement 2 is NOT correct.**
- The eye of the tropical cyclone is surrounded by a zone of strong winds with clouds extending vertically. This area is called 'wall cloud' zone and is the area of maximum winds. **Hence, statement 3 is correct.**

64. (d)

Explanation:

Explanation:

- Population and decadal change by residence from 2001 to 2011 was highest (55.9%) in the Indian union territory of Dadra and Nagar Haveli. **Hence, statement 1 is NOT correct.**



- Population and decadal change by residence from 2001 to 2011 was negative (−0.6%) in the state of Nagaland. **Hence, statement 2 is NOT correct.**

65. (b)

Explanation:

- In 1999, the Reserve Bank of India had set up a **Regulations Review Authority (RRA)** for reviewing the regulations, circulars, reporting systems, based on the feedback from the public, banks, and financial institutions. **Hence, statement 1 is NOT correct.**
- It will focus on **streamlining regulatory instructions**, reducing compliance burden of the regulated entities by simplifying procedures and reducing reporting requirements, wherever possible. **Hence, statement 2 is correct.**
 - It will also obtain **feedback from regulated entities**.
 - Regulated entities include commercial banks, urban co-operative banks, Non-Banking Financial Companies.

66. (c)

Explanation:

- The historical city of Kanchipuram is known for its spirituality, serenity, and silk.
 - It is situated on the banks of **River Vegavathi** (and not river Kaveri). **Hence, statement 1 is NOT correct.**
 - River Vegavathi is a tributary of the river Swarnamukhi.
- This historical city once had **1,000 temples**, of which only **126** (108 Shaiva and 18 Vaishnava) now remain.
- Its rich legacy has been the endowment of the **Pallava dynasty**, which made the region its **capital** between the 6th and 7th centuries and lavished upon its architectural gems that are a fine example of Dravidian styles. **Hence, statement 2 is correct.**
- The temples of Kanchipuram along with five other Indian places have been added to the tentative list of UNESCO's world heritage sites (it was not declared as a UNESCO World Heritage Site in 2011). **Hence, statement 3 is NOT correct.**
 - As per Operational Guidelines, 2019 of UNESCO, it is mandatory to put any monument/site on the Tentative List (TL) for one year before it is considered for the final nomination dossier.

- The other five places include; Satpura Tiger Reserve (Madhya Pradesh), Ghats of Varanasi (Uttar Pradesh), Megalithic Site of Hirebenkal (Karnataka), Maratha Military Architecture in Maharashtra and Bhedaghat-Lametaghat in Narmada Valley- Jabalpur (Madhya Pradesh).

67. (a)

Explanation:

Truck Farming

- The regions where farmers specialise in **vegetables only**, the farming is known as truck farming. **Hence, statement 1 is NOT correct.**
- The distance of truck farms from the market is governed by the distance that a truck can cover overnight, hence the name truck farming.

Collective Farming

- The basic principle behind this type of farming is based on social ownership of the means of production and collective labour.
- Collective farming or the **model of Kolkhoz** was introduced in the erstwhile **Soviet Union** to improve upon the inefficiency of the previous methods of agriculture and to boost agricultural production for self-sufficiency. **Hence, statement 2 is correct.**

Footloose Industries

- Footloose industries can be located in a **wide variety of places**. They are **not dependent on any specific raw material**, weight losing or otherwise. **Hence, statement 3 is correct.**
- They largely depend on component parts which can be obtained anywhere.
- They produce in small quantities and also employ a small labour force.
- These are generally not polluting industries. The important factor in their location is accessibility by road network.

68. (b)

Explanation:

Type of Climate Areas:

- | | |
|----------------------------------|------------------------------------|
| 1. Monsoon with short dry season | : West coast of India south of Goa |
| 2. Monsoon with dry summer | : Coromandel coast of Tamil Nadu |



3. **Tropical Savannah** : Most of the Peninsular plateaus, south of the Tropic of Cancer
4. **Semi-arid Steppe** : North-western Gujarat, some parts of western Rajasthan and Punjab
5. **Hot desert Extreme** : Western Rajasthan
6. **Monsoon with dry winter** : Ganga plain, eastern Rajasthan, northern Madhya Pradesh, most of North-east India
7. **Cold humid winter with short summer** : Arunachal Pradesh
8. **Polar type** : Jammu and Kashmir, Himachal Pradesh and Uttarakhand

Hence, only pair 3 is correctly matched.

69. (c)

Explanation:

- **Bishnoi Movement** is a religious sect found in the Western Thar Desert and northern states of India. It was founded by Guru Maharaj Jambaji in 1485 AD in the Marwar (Jodhpur) desert region of **western Rajasthan**. Hence, pair 1 is correctly matched.
 - It is a non-violent community of nature worshippers. This movement was started by sage Sombaji around 1700 AD against deforestation. After that Amrita Devi forwarded the movement.
- **Chipko Movement** was launched from Gopeshwar in **Chamoli district, Uttarakhand** in 1973 (then Uttar Pradesh). The movement was to prevent the illegal cutting of trees in the Himalayan region (Uttarakhand). Hence, pair 3 is correctly matched.
 - Sunderlal Bahuguna and Chandi Prasad Bhatt were the leaders of this movement. The most notable characteristics of this movement were the involvement of women.
- **Appiko Movement** was launched in 1983, on the lines of the Chipko Movement, Pandurang Hegde launched a movement which came to be known as **Appiko Movement in Karnataka**. Hence, pair 2 is NOT correctly matched.

- Its main objectives were afforestation as well as development, conservation and proper utilization of forests in the best manner. The meaning of “appiko” is to express one's affection for a tree by embracing it.

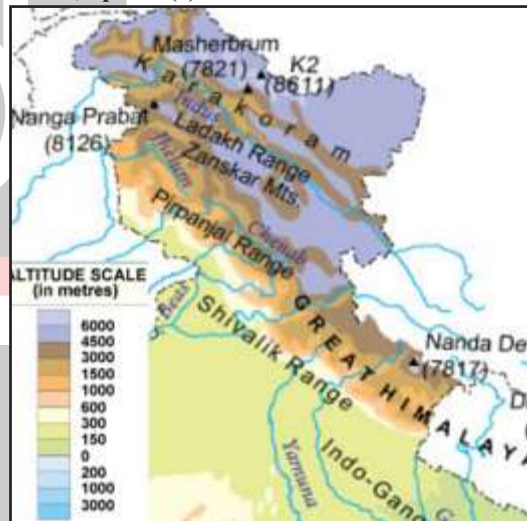
- **Jungle Bachao Andolan** was launched by the tribal community of **Singbhum district of Jharkhand** (then Bihar) agitating against the forest policy of the Government in 1982. Hence, pair 4 is NOT correctly matched.

- The Government wanted to replace the natural soil, forests with the high-priced teak.

70. (c)

Explanation:

In the North Western Himalayas, the North to South Order of Ranges is: **Karakoram, Ladakh, Zaskar, Pir Panjal**. Hence, option (c) is correct.



71. (c)

Explanation:

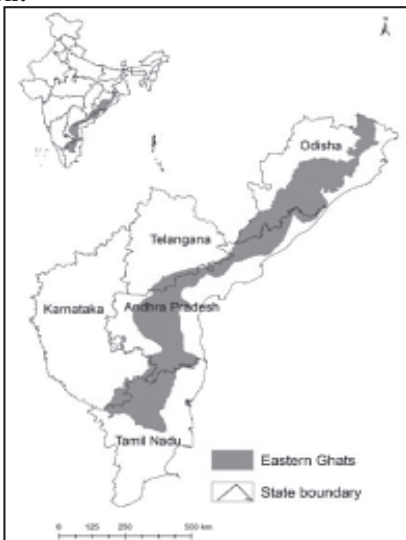
- Recently, the **International Energy Agency (IEA)** has released the **India Energy Outlook 2021** Report which explores the opportunities and challenges ahead for India as it seeks to ensure reliable, affordable and sustainable energy for a growing population.
- The India Energy Outlook 2021 is a new special report from the IEA's World Energy Outlook series.
 - The World Energy Outlook Report is released by the IEA annually.

Hence, option (c) is correct.



72. (b)

Explanation:



- The Eastern Ghats encompass the states of Odisha, Andhra Pradesh, Telangana, Karnataka, Tamil Nadu. Hence, option (b) is correct.

73. (a)

Explanation:

- Van Gujjars are the forest-dwelling nomadic community inhabiting the foothills of Himalayan States such as Uttarakhand, Himachal Pradesh and Jammu and Kashmir. Hence, statement 1 is NOT correct.
 - Usually, they migrate to the bugyals (grasslands) located in the upper Himalayas with their buffaloes and return only at the end of monsoons to their makeshift huts, deras, in the foothills.
 - They traditionally practice buffalo husbandry; a family owns up to 25 heads of buffaloes. They rely on buffaloes for milk, which gets them a good price in Uttarakhand and Uttar Pradesh markets.
- The Van Gujjars pursue seasonal migration from the Terai-Bhabar and Siwalik region of Uttarakhand to the higher bugyals in the Western Himalayas in summer and vice versa in winter. Hence, statement 2 is correct.

74. (a)

Explanation:

Majuli

- The Majuli Island sustains bio-diversity due to its geographical and physiographic conditions with climatic variations.

- The island is bound by the river Subarnsiri and her tributaries Ranganadi, Dikrong, Dubla, Chici and Tuni etc. on the north-west, the Kherkutia Suti (a spill channel of the river Brahmaputra) in the north-east and the main Brahmaputra River on the south and the south-west. Hence, option (a) is correct.
- Majuli falls within the greater Brahmaputra valley of Assam and is influenced by monsoon climate.

75. (c)

Explanation:

Ramsar Lakes in India:

- **Lonar Crater Lake** in Buldhana district of Maharashtra is a National Geo heritage Monument of India. The lake is both saline and alkaline in nature and is famous for a wide range of plant and animal life.
- **Chilika Lake** is brackish water lagoon in Odisha state it is the largest coastal lagoon in India and Asia's biggest saltwater lake. It is home to a number of migratory birds coming to India and the first Indian wetland of international importance.
- **Kolleru Lake** - Kolleru Lake is one of the largest freshwater lakes in India located in state of Andhra Pradesh and forms the largest shallow freshwater lake in Asia, 15 kilometres away from the city of Eluru. Kolleru is located between Krishna and Godavari deltas. Kolleru spans into two districts - Krishna and West Godavari. Hence, option (c) is correct.
- **Pangong Tso** of Ladakh is a saline water lake, situated at a height of about 4,350 m in the Himalayas. The lake is an important breeding ground for a variety of birds.

76. (c)

Explanation:

- A pathogen is a biological agent that causes disease or illness. Zoonotic pathogen refers to pathogens naturally transmitted between animals and humans. Hence, statement 1 is correct.
- **Types of Pathogens:**
 - Virus: made up of a piece of genetic code, such as DNA or RNA, and protected by a coating of protein.
 - Diseases caused by viruses: Chickenpox, Flu (influenza), Covid-19, Human immunodeficiency virus (HIV/AIDS), Mumps, measles and rubella.



- **Bacteria:** microorganisms made of a single cell. They are very diverse, have a variety of shapes and features, and have the ability to live in just about any environment, including in and on the body.
 - Diseases caused by bacteria: Cholera, Leprosy, Tuberculosis, Plague, Syphilis, Anthrax etc. **Hence, statement 2 is correct.**
- **Fungi:** They cause infection when they overgrow.
 - Examples of Fungal Infection: Mucormycosis, White Fungus, Yellow Fungus.
- **Parasites:** organisms that behave like tiny animals, living in or on a host and feeding from or at the expense of the host.
 - Diseases caused by Parasite: Malaria, African trypanosomiasis, babesiosis, leishmaniasis, and toxoplasmosis etc.
- The World Health Organization (WHO) and Switzerland have launched a BioHub facility that will allow pathogens sharing between laboratories and facilitate “analysis and preparedness” against them (and not G20 countries). **Hence, statement 3 is NOT correct.**

77. (b)

Explanation:

- Garo, Khasi and Jaintia hills are part of the peninsular plateau. **Hence, statement 1 is correct.**
- Diphu Pass is a mountain pass around the area of the disputed tripoint borders of India, China, and Myanmar. **Hence, statement 2 is correct.**
- Anthracite is the best quality of coal which carries 80 to 95% carbon content. It ignites slowly with a blue flame. It has the highest calorific value. It is found in small quantities in Jammu and Kashmir. **Hence, statement 3 is correct.**

78. (c)

Explanation:

White Fungus:

- White Fungus or Candidiasis is a fungal infection caused by a yeast (a type of fungus) called Candida. **Hence, statement 1 is correct.**
- Candida normally lives on the skin and inside the body, in places such as the mouth, throat, gut, and vagina, without causing any problems. **Hence, statement 2 is correct.**

- Candida can cause infections if it grows out of control or if it enters deep into the body (for example, the bloodstream or internal organs like the kidney, heart, or brain).

Black fungus:

- Black fungus is a serious but rare fungal infection caused by a group of moulds called mucormycetes, which is abundant in the environment.

Fungus:

- These are organisms separate from the plants and animal kingdoms.
- They are ubiquitous and are found in the soil, plants, decaying organic matter, water, air, damp places, and also in humans and animals.
- They play a very important role in our ecosystem along with bacteria, by degrading organic matter into simpler forms for the consumption of plants.
- They include household yeast, moulds, mushrooms, and several others.
- There are about 1,44,000 species of fungi (plural for fungus), out of which some of them are pathogenic to humans.
- The most common are Candida, Aspergillus, Cryptococcus, Histoplasma, Pneumocystis, and Mucormycetes.

79. (c)

Explanation:

Retreating Monsoon

- The months of October and November are known for retreating monsoons.
- By the end of September, the southwest monsoon becomes weak as the low pressure trough of the Ganga plain starts moving southward in response to the southward march of the sun.
- The monsoon retreats from the western Rajasthan by the first week of September. It withdraws from Rajasthan, Gujarat, Western Ganga plain and the Central Highlands by the end of the month.
- By the beginning of October, the low pressure covers northern parts of the Bay of Bengal and by early November, it moves over Karnataka and Tamil Nadu. By the middle of December, the centre of low pressure is completely removed from the Peninsula.
- **The retreating southwest monsoon season is marked by clear skies and rise in temperature.** The land is still moist. Owing to the conditions of high temperature and



humidity, the weather becomes rather oppressive. This is commonly known as the 'October heat'. Hence, **statement 1 is correct.**

- In the second half of October, the mercury begins to fall rapidly, particularly in northern India.
- The weather in the retreating monsoon is dry in north India but it is associated with rain in the eastern part of the Peninsula.
- Here, October and November are the rainiest months of the year.
- A bulk of the rainfall of the Coromandel coast is derived from these depressions and cyclones.

Western Cyclonic Disturbances

- The western cyclonic disturbances enter the Indian subcontinent from the west and the northwest during the winter months, originate over the Mediterranean Sea and are brought into India by the westerly jet stream.
- An increase in the prevailing night temperature generally indicates an advance in the arrival of these cyclones disturbances. Hence, **statement 2 is correct.**
- Inflow of western cyclones generally known as disturbances during the winter season and tropical depressions during the south-west monsoon period into India, creating weather conditions favourable to rainfall.
- The rainfall occurs in Punjab, Haryana, Delhi and western Uttar Pradesh. Although the amount is meagre, it is highly beneficial for rabi crops. The precipitation is in the form of snowfall in the lower Himalayas.

80. (c)

Explanation:

- Lunar eclipse takes place when the Moon moves into the Earth's shadow. The Earth has to be directly between the Sun and the Moon, and a lunar eclipse can only take place during a full Moon.
- First, the Moon moves into the penumbra – the part of the Earth's shadow where not all of the light from the Sun is blocked out. Part of the Moon's disc will look dimmer than a regular full Moon. Hence, **statement 1 is NOT correct.**
- And then the Moon moves into the Earth's umbra, where direct light from the Sun is totally blocked out by the Earth. This means the only light reflecting off the Moon's disc has already been refracted, or bent, by the Earth's atmosphere. Hence, **statement 2 is NOT correct.**

81. (d)

Explanation:

- **Culturable Wasteland:** Any land which is left fallow (uncultivated) for more than five years is included in this category. Hence, **statement 1 is NOT correct.**
 - It can be brought under cultivation after improving it through reclamation practices.
- **Current Fallow:** This is the land which is left without cultivation for one or less than one agricultural year.
 - Fallowing is a cultural practice adopted for giving the land rest. The land recoups the lost fertility through natural processes.
- **Fallow other than Current Fallow:** This is also a cultivable land which is left uncultivated for more than a year but less than five years. Hence, **statement 2 is NOT correct.**
 - If the land is left uncultivated for more than five years, it would be categorised as culturable wasteland.

82. (a)

Explanation:

- The Narmada is the largest west flowing river of peninsular India. It originates from Amarkantak and flows through the rift valley between Vindhya and Satpura ranges. It flows through the states of Madhya Pradesh, Maharashtra and Gujarat. Hence, **statement 1 is correct.**
- The Godavari is the largest river basin of peninsular India not Cauvery and extends over ten per cent of the total geographical area of the country. Hence, **statement 2 is NOT correct.**
- Godavari river basin is shared by Odisha, Madhya Pradesh, Chhattisgarh, Karnataka and Maharashtra.

83. (c)

Explanation:

- Mid-Day-Meal Scheme is a centrally sponsored scheme which was launched in 1995. Hence, **statement 1 is correct.**
- It is considered as the world's largest school feeding programme aimed to attain the goal of universalization of primary education. Hence, **statement 2 is correct.**
- Nodal Ministry for the implementation of the scheme is the Ministry of Education. Hence, **statement 3 is NOT correct.**



84. (d)

Explanation:

Rice

- Rice is a crop of tropical humid areas.
- India has about 3,000 varieties which are grown in different agro-climatic regions.
- These are successfully grown from sea level to about 2,000 m altitude and from humid areas in eastern India to dry but irrigated areas of Punjab, Haryana, western U.P. and northern Rajasthan.
- In southern states and West Bengal the climatic conditions allow the cultivation of two or three crops of rice in an agricultural year.
- In West Bengal farmers grow three crops of rice called 'aus', 'aman' and 'boro'.
- In the Himalayas and northwestern parts of the country, it is grown as a kharif crop during southwest Monsoon season.
- India contributes 21.6% of rice production in the world and ranked second after China in 2016.
- About one-fourth of the total cropped area in the country is under rice cultivation.
- The yield level of rice is high in Punjab, Tamil Nadu, Haryana, Andhra Pradesh, Telangana, West Bengal and Kerala. In the first four of these states almost the entire land under rice cultivation is irrigated.
- Punjab and Haryana are not traditional rice growing areas.
- Rice cultivation in the irrigated areas of Punjab and Haryana was introduced in the 1970s following the Green Revolution.
- **Genetically improved varieties of seed**, relatively high **usage of fertilizers and pesticides** and lower levels of susceptibility of the crop to pests due to **dry climatic conditions** are responsible for higher yield of rice in this region. **Hence, statement 1, 2 and 3 are correct.**

85. (c)

Explanation:

The Geological Survey of India has classified rock systems of the country into 4 major divisions:

1. The Archaean Rock System
2. The Purana Rock System
3. The Dravidian Rock System
4. The Aryan Rock System

1. **The Archaean Rock System:** The earliest phase of tectonic evolution was marked by the cooling and solidification of the upper crust of the earth surface in the Archaean era which is represented by the exposure of gneisses and granites especially on the Peninsula.

- a. The Archaean group of rocks consists of two systems-

i. **Achaean System:**

1. This is the oldest rock system of Indian subcontinent dating back to about 4 billion years .
2. They are found in the Aravalli hills, Deccan plateau and the northeast of India. They also occur in the roots of the mountain peaks all along the Greater Himalayas, Trans Himalayan ranges of Zaskar, Ladakh and Karakoram.
3. Because of their volcanic origin, they are crystalline and consist of sheet-like layers (foliated). These rocks are abundant in metallic as well as non-metallic minerals such as iron, manganese, copper, bauxite, gold, lead, mica, graphite , hornblende etc

ii. **Dharwar System:**

1. Formed between 2.5 billion to 1.8 billion year, these are the oldest metamorphic rocks of India. They are formed due to the metamorphosis of the sediments formed out of the Archean rocks.
2. They are rich in metallic minerals such as iron, manganese, gold, copper, etc. They are considered to be the storehouses of metallic minerals and hence have a high economic significance. **Hence, statement 2 is correct.**
3. Since they were first studied in the Dharwar region of Karnataka, they were named so. They are mainly found in the Aravallis, Chotanagpur plateau, Meghalaya and Tamil Nadu. **Hence, statement 1 is correct.**

2. **Purana Rock System:**

- The Cuddapah and Vindhyan rock systems are together known as the Purana rock system. They are formed by the erosion and deposition of Archean and Dharwar rocks, the process is believed to have taken place between 1400-600 million years ago. They are mostly sedimentary in nature.



a. **The Cuddapah Rock system:**

- i. Named after the Cuddapah (now Kedara) district of Andhra Pradesh.
- ii. It spread in crescent shape, approx 22000 sq km area of Andhra, Chhattisgarh, Maharashtra, Tamil Nadu, Delhi, Rajasthan, and lesser Himalayas.
- iii. They were formed when sedimentary rocks like sandstone, limestone etc., and clay were deposited in synclinal folds (between two mountain ranges).
- iv. They are rich in shale, slate, quartzite, iron ore (Inferior quality), manganese, asbestos, copper, nickel, cobalt, marble, jasper, building stones, and stones however they were of low quality.

b. **The Vindhyan rock system (1300-600 million years)**

- i. The system derives its name from the Vindhya mountains.
- ii. It extends from Rajasthan to Bihar forming a saucer shape.
- iii. The system comprises ancient sedimentary rocks superimposed on the Archaean base.
- iv. Mostly Unfossiliferous rocks and large areas of this belt is covered by the Deccan trap.
- v. It is devoid of metalliferous minerals but provides large quantities of durable stones, ornamental stones, limestone, pure glass making sand, etc.
- vi. The Vindhyan system has diamond bearing regions from which Panna and Golconda diamonds have been mined.

3. **The Dravidian Rock System:**

- a. The Paleozoic rock formation is known as Dravidian systems in India; during the Paleozoic era i.e., 600-300 million years ago.
- b. It is also known as the Carboniferous rocks System due to high quality Coal formation in the world.
- c. These rocks are mostly found in the extra-Peninsular regions of the Himalayas and the Gangetic plain and are very little in the Peninsular shield (Umaria in Rewa). PirPanjal, Handwara, Lida valley,

Annatnag of Kashmir, Spiti, Kangra & Shimla region of Himachal Pradesh Garhwal & Kumaon of Uttarakhand are the major regions of Dravidian rocks.

- d. The Dravidian rocks mainly include shales, sandstones, clays, quartzites, slates, salts, talc, dolomite, marble, etc.

4. **The Aryan Rock System:** The beginning of Upper carboniferous period is known as the Aryan Group.

- a. **The Tertiary:** These rocks belong to the Cenozoic era, formed around 60 million years ago. Characterized by two events- the final breaking up the old Gondwanaland and the upliftment of Tethys geosynclines or Himalayas.

b. **Deccan Trap:**

- i. Deccan Trap was formed when the Indian plate came over the Reunion hotspot in the Indian Ocean while travelling north towards the Eurasian plate, after breaking up from the Gondwana plate.
- ii. The continuous outpouring of magma, from the fissures over the Indian plate led to the formation of a layered structure called the Deccan Traps.
- iii. Basaltic lava covered an area of 5 lakh sq.km. encompassing the regions like Kutchch, Saurashtra, Maharashtra, Malwa plateau and north Karnataka.
- iv. The thickness of the Deccan traps decreases from west to east, with around 3000m in the west to just 150m in the east, and around 800m in the south.

c. **The Gondwana System:**

- i. The Gondwana rock system is named after the Gond tribe (indigenous people mainly found in the Telangana and Andhra Pradesh regions).
- ii. These are sedimentary deposits which were accumulated in the synclinal troughs on an older plateau surface.
- iii. This system contains huge carbon deposits within them. This makes them the largest source of coal in India, containing up to 98 percent of our coal deposits.



- iv. The Gondwana rocks are mainly found in Ranigunj, Jharia regions of Jharkhand, Damodar valley, Pench valley in Chhattisgarh and Madhya Pradesh, Godavari valley in Telangana and the Rajmahal hills of West Bengal.

86. (b)

Explanation:

Pradhan Mantri Swasthya Suraksha Yojana (PMSSY)

- PMSSY was announced in 2003 with the objectives of correcting regional imbalances in the availability of affordable/reliable tertiary healthcare services and also augmenting facilities for quality medical education in the country. **Hence, option (b) is correct.**
- It comes under the Ministry of Health and Family Welfare.
- It has two Components:
 - Setting up AIIMS like institutions.
 - Upgradation of government medical colleges in various states.
 - The project cost for the up-gradation of each medical college institution is shared by the Centre and the state.

87. (d)

Explanation:

Bauxite

- Bauxite is the ore, which is used in manufacturing aluminium. Aluminum is ideal for aircraft manufacture because it's lightweight and strong. **Hence, statement 2 is correct.**
- Bauxite is found mainly in tertiary deposits and is associated with **laterite rocks** occurring extensively either on the plateau or hill ranges of peninsular India and also in the coastal tracts of the country. **Hence, statement 1 is correct.**
- Odisha happens to be the largest producer of Bauxite. Kalahandi and Sambalpur are the leading producers. **Hence, statement 3 is correct.**
- The other two areas which have been increasing their production are Bolangir and Koraput.
- The patlands of Lohardaga in Jharkhand have rich deposits. Gujarat, Chhattisgarh, Madhya Pradesh and Maharashtra are other major producers. Bhavanagar, and Jamnagar in Gujarat have the major deposits.

- Chhattisgarh has bauxite deposits in Amarkantak plateau while Katni- Jabalpur area and Balaghat in M.P. have important deposits of bauxite. Kolaba, Thane, Ratnagiri, Satara, Pune and Kolhapur in Maharashtra are important producers. Tamil Nadu, Karnataka and Goa are minor producers of bauxite.

88. (c)

Explanation:

- The Non-Aligned Movement (NAM) was formed during the Cold War (1945-1991) as an organisation of States that **did not seek to formally align themselves with either the USA (Capitalism) or the Soviet Union (Socialism)** but sought to remain independent or neutral. **Hence, statement 1 is correct.**
- It was the **Bandung Conference**, held in 1955 that resulted in the adoption of the founding principles of the Non-Aligned Movement (NAM) (and **not the Havana Declaration**). **Hence, statement 2 is NOT correct.**
 - The purpose of the NAM was enumerated in the Havana Declaration of 1979 to ensure "the national independence, sovereignty, territorial integrity and security of non-aligned countries".
- It was held under the leadership of Josip Broz Tito of Yugoslavia, Gamal Abdel Nasser of Egypt, Jawaharlal Nehru of India, Kwame Nkrumah of Ghana, and Sukarno of Indonesia. **Hence, statement 3 is correct.**

89. (a)

Explanation:

- Hail is precipitation in the form of ice pellets. Thunderstorms producing hail are known as hail storms.
- During the period from March to May thunderstorms over northern India, particularly the sub-mountain areas are accompanied by hail. **Hence, statement 1 is correct.**
- Hail storms also occur due to Western disturbances during the winter months from December to February. **Hence, statement 2 is NOT correct.**
- However, hail is rare in the Southwest Monsoon period.

90. (b)

Explanation:

The tropic of Cancer lies at 23.5 degrees North.

The latitudes of cities mentioned are:

1. Gandhinagar- 23.21° N, 72.63° E



2. Ranchi- 23.34° N, 85.30° E **Hence, option (b) is correct.**
3. Bhopal- 23.25° N, 77.41° E
4. Kolkata- 22.57° N, 88.36° E



91. (c)

Explanation:

- The World Press Freedom Day (WPDF) is an annual celebration of press freedom, observed on 3rd May every year. The **main celebration is organized by the UN Educational, Scientific and Cultural Organization (UNESCO)**. **Hence, statement 1 is correct.**
 - The theme for WPDF 2021 is "Information as a Public Good".
- The day was proclaimed by the UN General Assembly in 1993, following the recommendation of UNESCO's General Conference in 1991.
- The day also marks the 1991 Windhoek Declaration (adopted by UNESCO). It is considered a benchmark for ensuring press freedom around the world.
 - **The Windhoek Declaration is a statement of press freedom principles put together by African newspaper journalists in 1991. Hence, statement 2 is correct.**
 - The Declaration calls for a free, independent and pluralistic media and is based on a strong belief in the connection between a fully independent press and a successful participatory democracy sector.

92. (c)

Explanation:

National Programme on Advanced Chemistry Cell Battery Storage:

- The Union Cabinet has approved an Rs. 18,100 crore Production Linked Incentive (PLI) scheme for manufacturers of Advanced Chemistry Cell (ACC) battery storage, to reduce imports.

- The scheme is called the National Programme on Advanced Chemistry Cell Battery Storage (NPACC).
- It is under the Ministry of Heavy Industries & Public Enterprises. **Hence, statement 1 is correct.**
- The plan is to set up a 50-gigawatt hour (GWh) manufacturing capacity for ACC batteries by attracting investments totalling Rs. 45,000 crore. **Hence, statement 2 is correct.**

Advanced Chemistry Cell (ACC):

- ACCs are the new generation of advanced storage technologies that can store electric energy either as electrochemical or as chemical energy and convert it back to electric energy as and when required.
- Such battery storage will cater not only to electric vehicles but also to the consumer electronics industry and electricity grids.

93. (b)

Explanation:

- Physiological density = total population / net cultivated area **Hence, statement 1 is NOT correct.**
- Agricultural density = total agricultural population / net cultivable area **Hence, statement 2 is correct.**
- Agricultural population includes cultivators and agricultural labourers and their family Members. **Hence, statement 3 is NOT correct.**
- Population Doubling Time: Population doubling time is the time taken by any population to double itself at its current annual growth rate. **Hence, statement 4 is NOT correct.**

94. (b)

Explanation:

- The National Small Savings Fund (NSSF) within the Public Account of India was established in 1999. **Hence, statement 1 is NOT correct.**
 - The objective of the creation of the NSSF was to de-link small savings transactions from the Consolidated Fund of India and ensure their operation in a transparent and self-sustaining manner.
 - Since NSSF operates in the public account, its transactions do not impact the fiscal deficit of the Centre directly.



- The Fund is administered by the Government of India, Ministry of Finance (Department of Economic Affairs) under National Small Savings Fund (Custody and Investment) Rules, 2001, framed by the President of India under Article 283 (1) of the Indian Constitution. **Hence, statement 2 is correct.**

95. (d)

Explanation:

The Northern Plains can be arranged from South to North in the following way:

- Southernmost is **Khader** which are fresh river deposits on new flood plains. They are known as Dhayas in Punjab, Khol in Uttar Pradesh, Bhils in West Bengal, and Taal in Bihar.
 - Then comes **Bangar** which are old flood plains of rivers and Drylands
 - Then comes the **Terai** Region which is marshy and swampy. Sal forests are found in this region.
 - The Northernmost is **Bhabar** which are alluvial fans of Himalayan rivers it consist of coarse deposits and large boulders, the river disappears in this region.
- Hence, option (d) is correct.**

96. (c)

Explanation:

- Neanderthal (*Homo neanderthalensis*, *Homo sapiens neanderthalensis*) is member of a group of archaic humans who emerged at least 2,00,000 years ago during the Pleistocene Epoch (about 2.6 million to 11,700 years ago). **Hence, statement 1 is correct.**
 - They were replaced or assimilated by early modern human populations (*Homo sapiens*) between 35,000 and perhaps 24,000 years ago.
- A genome is all the genetic matter in an organism. It is defined as an organism's complete set of Deoxyribose Nucleic Acid (DNA), including all of its genes. **Hence, statement 2 is correct.**
 - In humans, a copy of the entire genome contains more than 3 billion DNA base pairs.

97. (c)

Explanation:

Sunderlal Bahuguna (1927-2021):

- Initiated the Chipko Movement to guard the trees on the Himalayan slopes. **Hence, statement 1 is correct.**

- Known for coining the Chipko slogan 'ecology is the permanent economy' **Hence, statement 3 is correct.**
 - After the Chipko movement in the 1970s, he gave the message across the globe that ecology and ecosystem are more important. He was of the opinion that ecology and economy should go together.
- Campaigned against the Tehri Dam on River Bhagirathi, a mega-project with devastating consequences. He undertook one of India's longest fasts after Independence, over 56 days.
- Took 4,800 km Kashmir to Kohima padayatra (foot march) in the early 1980s to bring attention to the entire Himalayan region.
- Supported women-led movements against the liquor mafia in the hills, and for the Beej Bachao Andolan, a movement to save Himalayan agricultural biodiversity from being wiped out by the unsustainable, chemical-intensive Green Revolution.
- He was awarded the Padma Vibhushan in 2009. **Hence, statement 2 is correct.**

Chipko Movement:

- It was a non-violent agitation that originated in Uttar Pradesh's Chamoli district (now Uttarakhand) in 1973.
- The name of the movement 'chipko' comes from the word 'embrace', as the villagers hugged the trees and encircled them to prevent being hacked.
- It is best remembered for the collective mobilisation of women for the cause of preserving forests, which also brought about a change in attitude regarding their own status in society.
- Its biggest victory was making people aware of their rights to forests, and how grassroots activism can influence policy-making regarding ecology and shared natural resources.
 - It led to a ban on commercial felling of trees above 30 degrees slope and above 1,000 MSL (mean sea level) in 1981.

98. (b)

Explanation:

The Indus River





It is one of the **largest river basins of the world**, covering an area of 11,65,000 sq. km (in India it is 321, 289 sq. km and a total length of 2,880 km (in India 1,114 km). The Indus also known as the **Sindhu**, is the westernmost of the Himalayan rivers in India.

It originates from a glacier near **Bokhar Chu in the Tibetan region** at an altitude of 4,164 m in the Kailash Mountain range.

The Indus receives a number of Himalayan tributaries such as the Shyok, the Gilgit, the **Zaskar**, the **Hunza**, the Nubra, the Shigar, the **Gastering** and the Dras. **Hence, 1, 2 and 3 are correct.**

Note: Ravi river is a tributary of Chenab river.

99. (a)

Explanation:

- Amarkantak is a pilgrim town in Anuppur, Madhya Pradesh, India. The Amarkantak region is a unique natural heritage area and is the meeting point of the Vindhya and the Satpura Ranges, with the Maikal Hills being the fulcrum. This is where the Narmada River and the Son River emerge. **Hence, option (a) is correct.**

Extra Information:

- The town of Amarkantak is surrounded by a rich variety of flora with medicinal properties. Amarkantak falls on

the Kanha-Achanakmar Corridor, a hilly region covered with dense forests. Kanha and Achanakmar are both conserved wildlife parks in India famous for Bengal tigers. Achanakmar-Amarkantak Biosphere Reserve is one of the Ten biosphere reserves included in the World Network of Biosphere Reserves, based on the UNESCO Man and the Biosphere (MAB) Programme list.



100. (d)

Explanation:

- The System of Air Quality and Weather Forecasting And Research (SAFAR) is a national initiative introduced by the Ministry of Earth Sciences (MoES) to measure the air quality of a metropolitan city, by measuring the overall pollution level and the location-specific air quality of the city. **Hence, statement 1 is NOT correct.**
- The system is indigenously developed by the Indian Institute of Tropical Meteorology (IITM), Pune and is operationalized by the India Meteorological Department (IMD). **Hence, statement 2 is NOT correct.**
- The ultimate objective of the project is to increase awareness among the general public regarding the air quality in their city so that appropriate mitigation measures and systematic action can be taken up.
 - SAFAR is an integral part of India's first Air Quality Early Warning System operational in Delhi.
- It monitors all weather parameters like temperature, rainfall, humidity, wind speed, and wind direction, UV radiation, and solar radiation.
- Pollutants Monitored: PM2.5, PM10, Ozone, Carbon Monoxide (CO), Nitrogen Oxides (NOx), Sulfur Dioxide (SO2), Benzene, Toluene, Xylene, and Mercury. **Hence, statement 3 is NOT correct.**

